Flows & Liquidity

Has investors leveraged peaked?

Global Markets Strategy

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- **After reaching extreme levels earlier this year, there are currently signs of retreat in retail investors' leverage in both options and margin accounts, presenting a potential headwind for tech stocks going forward.**
- **Similarly, there are signs of retreat in risk parity fund leverage in recent weeks following a historical high in mid May.**
- **Signs of peaking in hedge fund or bank leverage look more tentative.**
- **Both corporate and household leverage have been on a declining trend since the pandemic, thus posing little vulnerability to macro shocks.**
- **A modest deterioration in the global bond supply-demand balance for 2026 is already priced in by bond markets.**

Cross Asset Fund Flow Monitor

Current level shows the latest percentile of weekly flows; Min is denoted by 0 and Max by 1. As of 17th June 26.

MF & ETF Flows	Min	Max	4 wk avg ($bn)	2025 avg ($bn)
All Equities			35.3	8.1
All Bonds			17.0	11.3
US Equities			35.9	3.5
US Bonds			9.3	4.3
Non-US Equities			-0.6	4.6
Non-US Bonds			7.7	7.0
US HG Bonds			2.0	3.4
US HY Bonds			0.6	0.4
US Lev. Loans *			0.6	0.0
US MMFs			0.8	4.0
EM Equities			-1.4	0.9
EM Bonds			0.81	0.50
Japan Equities			-2.1	-0.1
China Equities			-1.65	-0.15
Europe				
Europe Equities			-2.1	0.9
Europe Bonds			5.2	4.3
Europe HG Bonds			0.1	0.9
Europe HY Bonds			0.15	0.17
Europe MMFs			6.4	3.8
Other Equities			5.39	3.00

Source: Lipper, ICI, Bloomberg Finance L.P. and J.P. Morgan Flows & Liquidity.
* US LL historical flows are monthly averages converted to weekly for comparison. China onshore A-share ETFs have been excluded.

Cross Asset Positioning Monitor

Current level shows the latest percentile, Min is denoted by 0 and Max by 1.

As of 23-Jun-26	MIN	MAX	Current percentile
Equities			0.64
Govt Bonds			0.62
Credit			0.33
Dollar			0.73
Commodities ex Gold			0.38
Gold			0.43
Bitcoin			0.52
EM Equities			0.59
EM Bonds/FX			0.21
Japan Equities			0.88
Europe Equities			0.59
US Equity Sectors:			
Energy			0.62
Materials			0.59
Industrials			0.62
Discretionary			0.60
Staples			0.13
Health Care			0.43
Financials			0.17
Technology			0.66
Communication Services			0.36
Utilities			0.44

Source: J.P. Morgan Flows & Liquidity.
Cross Asset Positioning Monitor aggregates across the various position indicators of Appendix ranging from positioning proxies across various futures contracts, momentum signals as proxies of how trend-following funds/CTAs are positioned, mutual fund betas as proxies of how mutual fund managers are positioned, risk parity fund positioning and leverage proxies, hedge fund betas as proxies of how hedge fund managers are positioned, client surveys, asset allocation estimates of private non-bank investors at global level, short interest indicators, etc.

See page 28 for analyst certification and important disclosures.

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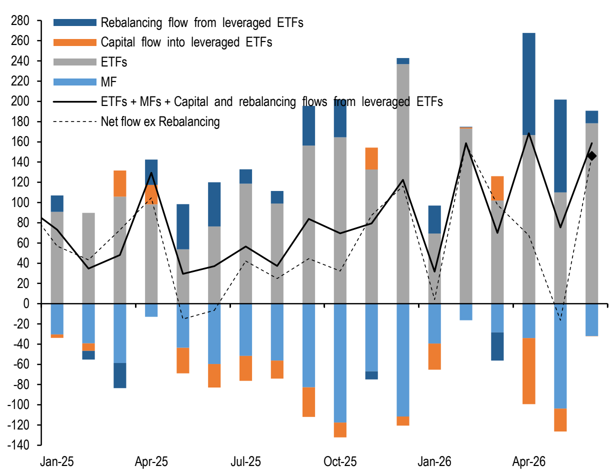
- We argued in our last publication that the rising trend in semiconductor stock volatility coupled with elevated semiconductor exposures in investor positioning are raising the risk of more frequent semiconductor "VaR shocks" from here. One manifestation of elevated investor positioning is leverage. How high is investors' leverage at the moment? And do we see signs of deleveraging?
- To answer these questions, we look at different forms of leverage across investor types. Starting with retail investors, we examine three main forms of their leverage: their flows into leveraged ETFs, their buying of options and their borrowing via margin accounts.
- We explained previously in our publication the increasing importance of the rebalancing flows emanating from leveraged ETFs ("*A $100bn rebalancing flow boosted April's equity rally*", May 6th 2026). These leveraged ETFs, with an AUM of $247bn globally, typically have a tech focus and have thus acted as a big amplification force for tech stocks in recent months, in particular in places like Korea. Indeed, Figure 1 and Figure 2 show how large these leveraged ETF rebalancing flows were in March, April and May. And by looking at the capital flow into leveraged equity ETFs (orange bars in Figure 1 and Figure 2), there is little evidence of retail investors retreating from these leveraged ETFs beyond the typical profit taking seen in strongly up months.

Figure 1: Flows stemming from equity funds globally

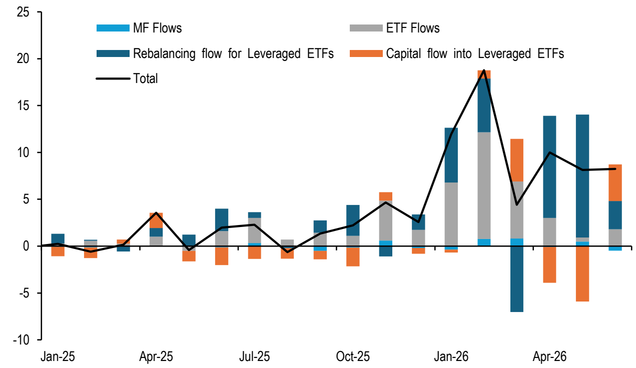
In $bn per month



Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

Figure 2: Flows stemming from equity funds that invest in Korea

In $bn per month.

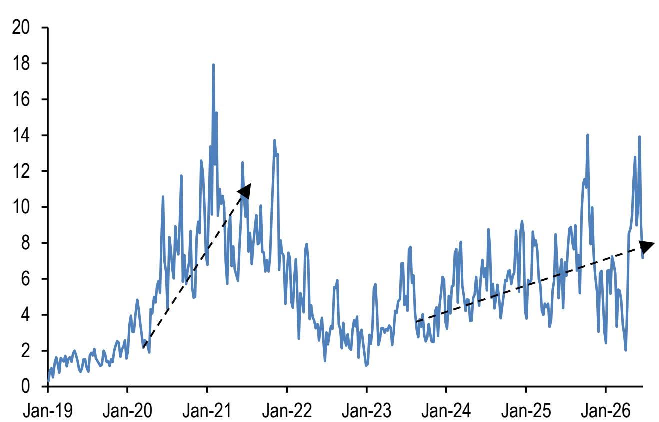


Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

- Where retail investors' leverage appears to be retreating is in options and margin accounts. In options, our proxy of retail investors' option buying based on OCC data (on exchange-traded call option buying by customers with less than 10 contracts), shows signs of retreat after peaking on June 5th (Figure 3). The peak on June 5th at close to 14 million contracts matches the previous peaks seen in October 2025 and November 2021. Following these previous peaks, tech stocks saw a multi-month correction with the bottom/capitulation in tech stocks coinciding with low readings in this metric of between 2-4 million contracts. In other words, the peaking of this metric since the high of June 5th suggests the retail impulse in the options space is currently subsiding and thus presents a potential headwind for tech stocks (the preferred habitat of retail investors) going forward.

Figure 3: Exchange-traded Call Option Buys at Open minus Sells at Open for Customers with less than 10 contracts for options on individual equities

In mn contracts. Last obs is for the week ending 19th June 2026.



Source: OCC, J.P. Morgan Flows & Liquidity.

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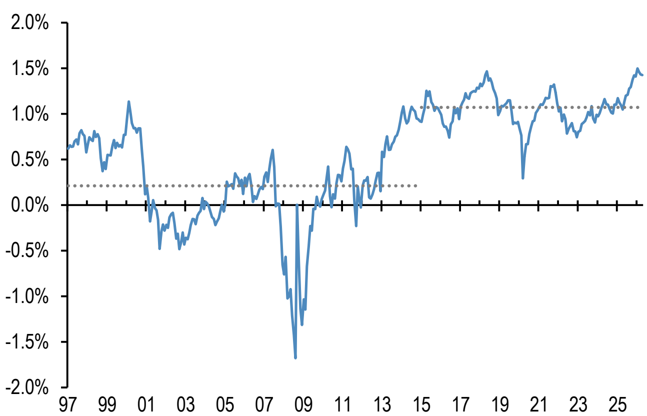
- For margin accounts, we typically use the NYSE Net Debit balance as a proxy for US individual investor leverage. While in principle it includes both retail and institutional investors, we believe the majority is likely retail investors. In contrast to hedge funds, who can get leverage more easily or cheaply via options and futures, individuals rely more on Federal Reserve Regulation T. This regulation, which governs customer cash accounts and the amount of credit that brokerage firms and dealers may extend to customers for the purchase of securities, stipulates that one can borrow up to 50% of the purchase price of securities that can be purchased on margin. The NYSE Net Debit balance is calculated as the difference between margin debit balance minus the sum of total credit balances (cash accounts credit + margin account credit). This leverage proxy suggests US individual investor leverage via margin accounts stands at extreme levels by historical standards with tentative signs of retreating in recent months (Figure 4). This year's peak matches that seen at the end of 2021 or before then in mid 2018, with both of those previous episodes having been followed by multi-month correction in equity markets presenting a potential headwind.

Figure 4: Net Debit balances in NYSE margin accounts

The NYSE margin account Net Debit balance is equal to the margin debit balance minus the sum of total credit balances (cash account credit +margin account credit). The blue line is constructed by the Net Debit balance in $bn divided by the market capitalisation of the S&P500 Index.

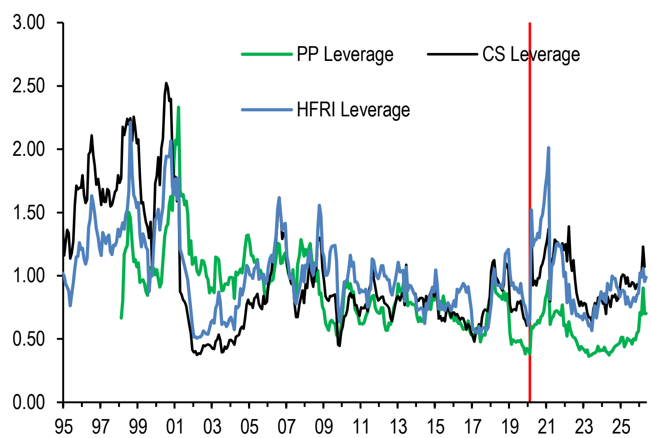


Source: FINRA, NYSE, J.P. Morgan Flows & Liquidity.

- In other words, after reaching extreme levels earlier this year, there are currently signs of retreat in retail investors' leverage in both options and margin accounts.
- What about hedge fund leverage? Figure 5 illustrates our proxy for aggregate hedge fund leverage, based on the ratio of hedge fund return volatility to the volatility of underlying asset returns, the rationale being that the higher the leverage the higher the volatility of hedge funds vs the volatility of the underlying assets. To construct our hedge fund leverage proxy, we divide hedge fund index return volatility by asset return volatility for each category. We use S&P 500 returns as a proxy for Equity Long/Short, Equity Short, and Equity Neutral funds and global bond index USD-hedged returns for Macro and Fixed-Income Arbitrage funds. The high yield corporate bond index, our proxy for hedge fund leverage, has been drifting higher since 2023 with March 2026 marking a potential local peak. While hedge fund leverage does not look particularly high by historical standards, especially compared to pre-2008 financial crisis norms, any further signs of peaking from here could add fuel to the idea that the rising trend in hedge fund leverage since 2023 is behind us.

Figure 5: Estimated HF Leverage

Our measure of HF leverage is a weighted average of the estimated leverage for five HFR hedge fund styles: Equity long/short, Equity short, Macro, Fixed Income arbitrage, Convertible arbitrage and Emerging Markets, Equity neutral and Eventdriven/ Distressed debt. For each style, we divide the hedge fund index return volatility by asset return volatility, which we proxy by S&P 500 returns for Equity long/short, Equity short and Equity neutral, Global bond index USD hedged returns for Macro and Fixed-Income arbitrage, high-yield returns for Convertible arbitrage and Event driven/Distressed debt and EMBIG returns for Emerging Markets. Last obs is Apr'26 for Pivotal Path, HFR & Credit Suisse.

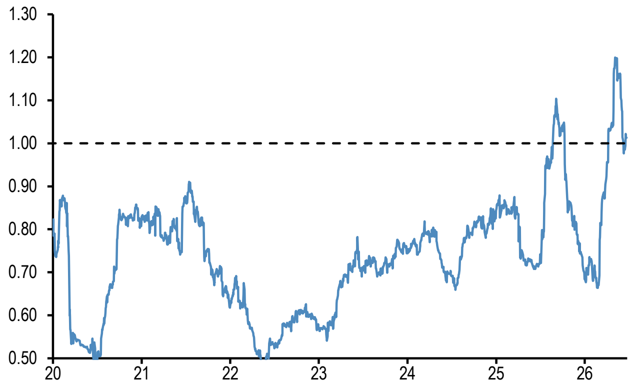


Source: Pivotal Path, Credit Suisse, HFR, J.P. Morgan.

- In risk parity fund space, there are more clear signs of peaking in their leverage, with our implied leverage proxy in Figure 6retreating in recent weeks after reaching its highest level in over a decade mid-May.

Figure 6: Implied leverage of risk-parity funds

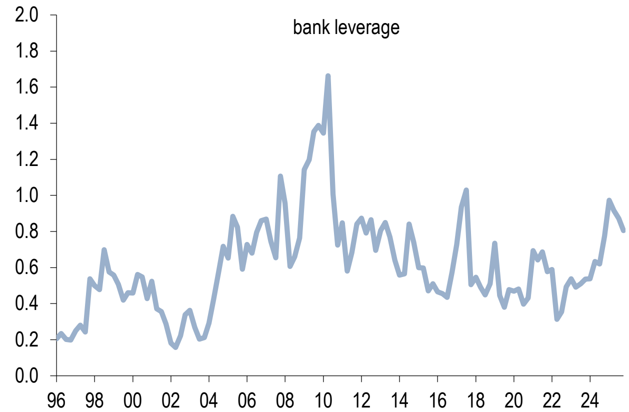
Ratio of 3-month rolling volatility of risk-parity funds' returns to the volatility of our risk parity strategy benchmark.



Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

Figure 7: Estimated US bank leverage

In the case of US banks, we proxy their leverage by the ratio of the volatility of their trading profits over the volatility of their assets. Quarterly data on the trading profits of US commercial banks are available from the Office of the Comptroller of the Currency (OCC).



Source: OCC, Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

- Similar to hedge funds, banks have also been raising their leverage in recent years due to deregulation. Regulations requiring banks to hold more capital and restrictions on proprietary trading by commercial banks meant bank leverage had declined after the financial crisis of 2008 and remained relatively contained until 2023. But a renewed deregulation drive by the US administration induced banks to increase their leverage. Despite the past year's increase, our bank leverage proxy remains well below pre-financial crisis norms with tentative signs of peaking (Figure 7).The bank leverage proxy in Figure 7 is proxied by the ratio of the volatility of bank trading profits over the volatility of their assets. Again, the rationale being that the higher the leverage the higher the volatility of bank trading profits vs the volatility of the underlying assets.

- What about economic leverage? Our broad metrics of economic leverage in Figure 8 suggests that, as economies have recovered from the pandemic shock, both corporate and household leverage have been on a declining trend. A relatively contained backdrop in terms of leverage, in turn, helps to reduce macro vulnerability to shocks. Moreover, as we have noted previously, despite the sharp rise in interest rates since early 2022, the net interest paid by non-financial corporates as a proportion of cash flows generated has declined markedly for US non-financial corporates (Figure 9), as the interest rates on short-term assets re-priced faster than the interest rates on their liabilities. Taken together, Figure 8 and Figure 9 suggest that neither corporate nor household leverage appear to be posing a vulnerability to macro shocks.

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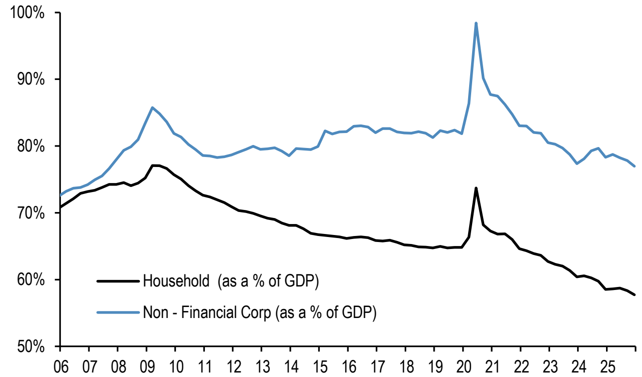
Figure 8: Stock of non-financial corporate sector & household debt as % of nominal GDP for Global GDP ex-China

Quarterly data, last obs. is Q4 2025.

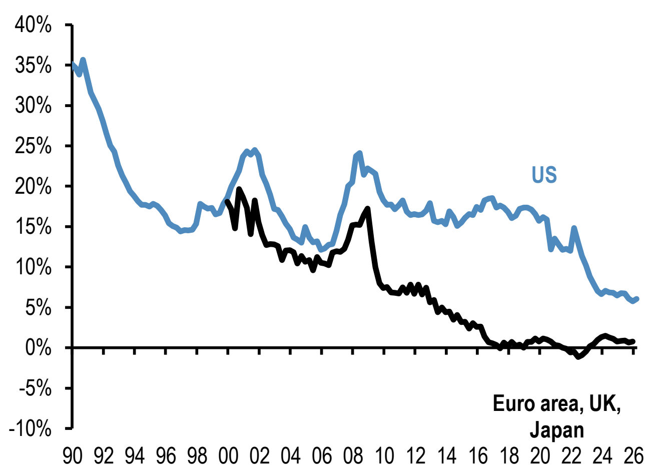


Source: National central banks, BIS, J.P. Morgan Flows & Liquidity.

- In all, after reaching extreme levels earlier this year, there are currently signs of retreat in retail investors' leverage in both options and margin accounts, thus presenting a potential headwind for tech stocks tech stocks going forward. Similarly, there are signs of retreat in risk parity fund leverage in recent weeks following a historical high in mid May. Signs of peaking in hedge fund or bank leverage look more tentative. Both corporate and household leverage have been on a declining trend since the pandemic, thus posing little vulnerability to macro shocks.

Figure 9: Net interest paid as % of non-financial corporate cash flows from operations

Till Q1'26 for US and G4 ex US



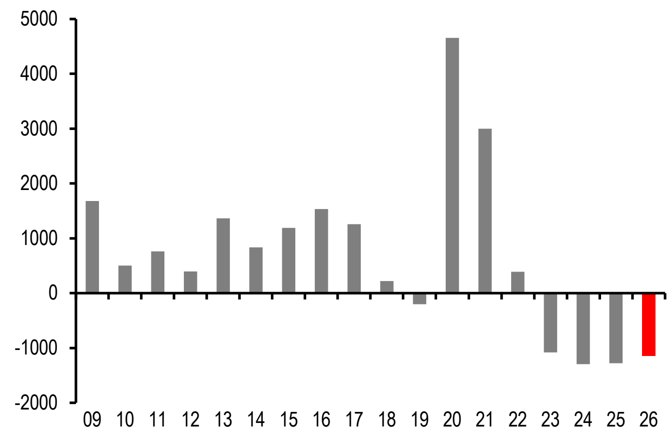
Source: Federal Reserve, ECB, BoJ, BoE, J.P. Morgan Flows & Liquidity.

A modest deterioration in the global bond supply-demand balance for 2026 is already priced in by bond markets

- With the recent release of the US flow of funds for 1Q26, quarterly worldwide fund flow data from the ICI covering 1Q26, as well as updated supply estimates from our colleagues' mid-year outlooks, we update our estimates for the global bond supply-demand balance (F&L, Nov 26th, 2025).
- Starting with demand, net bond demand by G4 central banks was broadly flat in 2025 vs. 2024 at just under -$1.3tr. With the Fed continuing to reinvest its maturing Treasuries across the curve, while MBS run-offs are reinvested in T-bills, this implies an ongoing modest negative duration impulse from the Fed (given we exclude T-bills from our global bond supply-demand analysis).For the ECB, both APP and PEPP holdings continue to run-off, and the BoE continues its QT via ongoing modest active sales as well as maturing bonds. And the BoJ has gradually accelerated its QT, and looks set to reduce its JGB holdings by around JPY48tr over the next 12 months. Net of these influences, we continue to G4 central bank net purchases at around -$1.2tr for 2026, or an improvement in demand of around $0.1tr for 2026 vs. 2025.

Figure 10: Net QE by G4 central banks

In $bn.



Source: Federal Reserve, ECB, BoE, BoJ, J.P. Morgan Flows & Liquidity.

- What about retail investors? In our 2026 global bond supply-demand analysis, we used a forecasting model that forecasts current year annual bond flows as a % of AUM as a function of the previous year's bond flow and bond returns. For this year, based on bond flows and returns in 2025, the model forecasted around $1.1tr of inflows into bond funds, which would have represented around a $0.2tr deterioration in bond demand from 2026 vs. 2025.

When we look at year-to-date inflows into bond funds, including quarterly reporting funds for 1Q and monthly or higher frequency for 2Q and add to this the average additional inflows into quarterly reporting funds since the start of 2025, this suggests that inflows into bond funds are, if anything, running at an annualized pace that is modestly stronger than the 2025 and 2024 pace of around $1.4tr. This would imply a modest improvement in in bond demand from retail investors in 2026 vs. 2025 of around $0.1tr.

- That said, as we have argued previously, the duration impulse of this support from retail investors in 2025 to-date has been fading. Figure 11 depicts this cumulative duration impulse of net flows into bond ETFs, and shows that inflows into bond ETFs have been directed into funds with shorter than average duration since late 2024.

Figure 11: Duration impulse from net inflows into bond ETFs

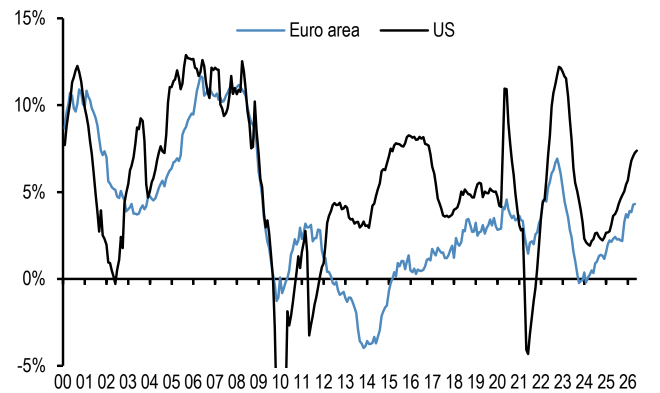
In $bn. Measured as the cumulative difference between daily net daily bond ETF flows adjusted to Bloomberg US Agg index duration equivalents and the bond ETF flows in dollar terms.

Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

- What about G4 commercial banks? In 2025, G4 commercial banks on aggregate bought around $1.1tr of bonds after around $0.9tr of purchases in 2024, or an improvement of around $0.2tr. We had previously estimated a somewhat larger improvement based on 2H25 data up to around October, but the realised pace was dragged down by net sales in December in particular. For 2026, we saw scope for bond demand to remain robust, as a resilient economic backdrop should support ongoing loan and deposit growth, which in turn should support some demand for bonds e.g. for liquidity and duration management purposes. Indeed, loan growth has continued to pick up in both the US and the Euro area (Figure 12), and in principle the ongoing ECB QT that continues to drain reserve would provide some ongoing support from banks replacing reserves with bonds to ensure they have sufficient liquid assets. That said, there are some signs that the share of debt securities of total assets among both US and Euro area banks has plateaued (Figure 13) after a relatively steady increase in recent years. And the 2026 to-date pace is now running at a $0.8tr annualized pace, implying a somewhat larger deterioration in bond demand of around $0.3tr in 2026 vs. 2025, compared to a $0.2tr deterioration that we had previously pencilled in.

Figure 12: Y/y loan growth for US and Euro area banks

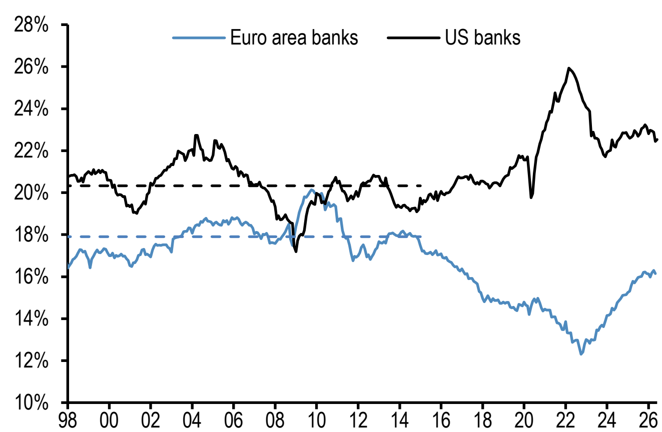
In %. Commercial bank loans and leases from the Fed's H.8 release for the US, ECB data on loans to non-financial corporations ex. general government for the Euro area.



Source: Fed, ECB, J.P. Morgan Flows & Liquidity.

Figure 13: US and Euro area banks' holdings of debt securities as a share of total assets

In %. Dashed lines are pre-ECB QE averages.



Source: Federal Reserve, ECB, J.P. Morgan Flows & Liquidity.

- What about foreign official demand? Unfortunately, the IMF's COFER data for 1Q26 have not yet been released as of the time of writing, but we can use our timelier proxy on EM FX reserve holdings for the first five months of the year, adjusting changes in reserves for currency and bond returns. This proxy suggests that, despite notable outflows in March, on net they have continued to

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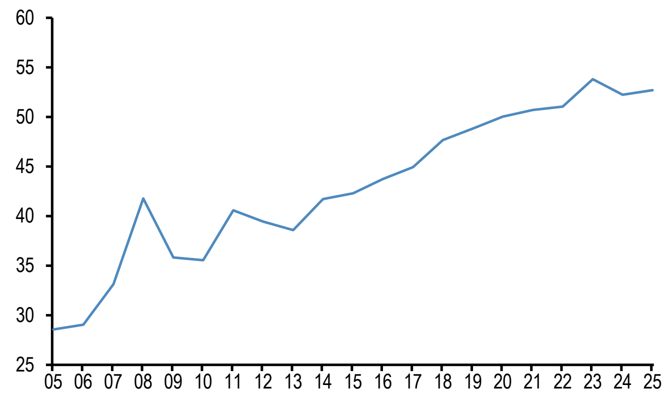
accumulate FX reserves YTD. As we noted previously, this is likely in part due to the fact that the disruption to oil flows from the Gulf region has not been uniform, meaning that the recycling of current account surpluses by oil producing countries have been perhaps less disrupted than feared, and in part as beneficiaries of the ongoing tech boom may have recycled some of increased current account surpluses via reserve accumulation (F&L, May 27th). That said, in 2H25 our proxy had over-predicted the reserve accumulation relative to subsequent COFER data. This means we keep our projection of an improvement in bond demand from reserve managers in 2026 vs. 2025 at $0.2tr, while acknowledging that the YTD pace of reserve accumulation from this proxy presents some upside risk.

- What about G4 pension funds and insurance companies? Based on updated flow of funds data for G4 economies for 2025, the overall net bond demand in reached around $850bn, modestly higher than to our previous projection. For 2026, we saw the backdrop as supportive for bond demand from G4 pension funds and insurance companies. In part, this reflects the fact that the largest 100 US private defined benefit pension funds captured in the US remain in surplus (see Chart A30 in the Appendix) as well as an ongoing rise in their bond allocations (Figure 14) as these funds are a mature universe that are largely closed to new entrants or in many cases to further accruals to existing members. In addition, if the improvement in the funded status of state and local funds continues (Figure 15), there could be an increased incentive for these funds to de-risk in coming years even if de-risking demand from more mature private defined benefit pension funds wanes. That said, while we had expected some pending rebalancing demand from the strength in equity markets in 2H25 to provide a further source of support this year, the released data for 2H suggests pace of bond purchases had already picked up in 2H. With less pending rebalancing from last year, and with some signs that the variation in bond demand around its quarterly average has been weaker than we had previously assumed (F&L, Jun 17th), we now see a modest deterioration in G4 pension fund and insurance company bond demand in 2026 vs. 2025 of around $0.1tr, vs. an improvement of $0.2tr previously.

Figure 14: Bond allocation of private defined benefit pension funds in the Milliman 100 index

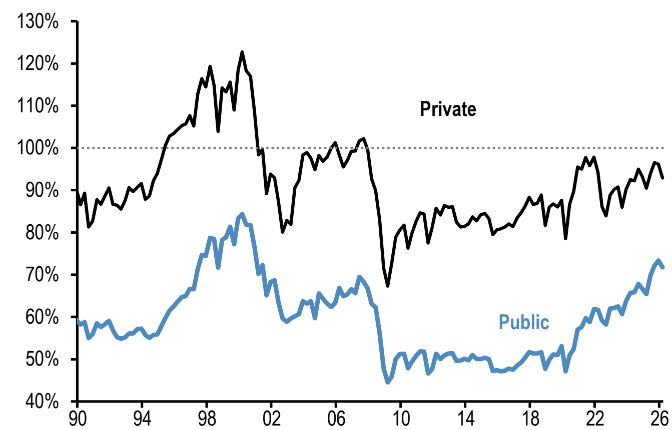
In %.



Source: Milliman, J.P. Morgan Flows & Liquidity.

Figure 15: Funded ratios of US private and public defined benefit pension funds

In %.



Source: Federal Reserve, J.P. Morgan Flows & Liquidity.

- Finally, what about bond supply? Taking updated net issuance projections from our strategists' 2026 mid-year outlooks, we estimate net supply for 2026 at around $4.9tr (Figure 16). This is little changed from our previous projection as a modest increase in spread product supply has been largely offset by a modest reduction on government supply.

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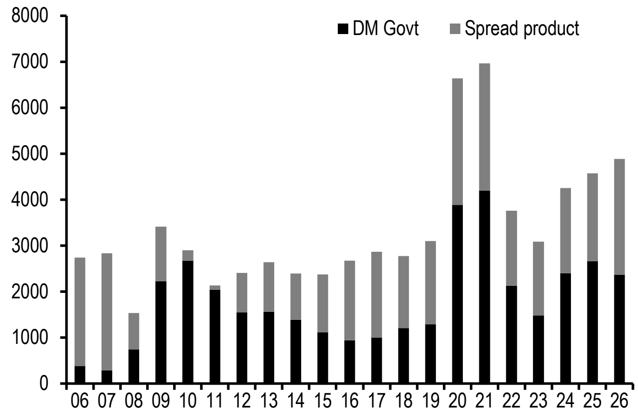
Figure 16: Net global bond supply

$bn per annum.

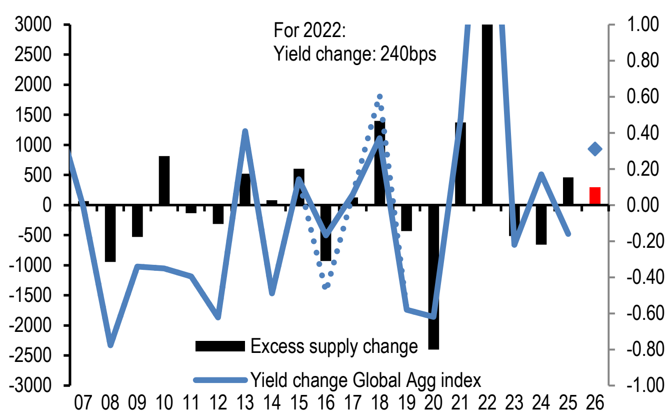


Source: J.P. Morgan Flows & Liquidity.

- Putting it all together, the above analysis now suggests little change in overall global bond demand in 2026 vs. 2025, vs. a $0.1tr improvement we had pencilled in previously. With an increase in bond supply of $0.3tr in 2026 vs. 2025, which now implies a deterioration in the supply-demand balance of around $0.3tr vs. $0.2tr previously. Based on the relationship between annual changes in excess supply and the Global Agg bond index yield, this would imply an increase in Global Agg index bond yield of around 15-20bp, somewhat less than the YTD increase in Global Agg yields.

Figure 17: Annual change in the balance between global bond supply and demand

Change in excess bond supply in $bn per annum in the left axis calculated as the difference between changes in global bond supply and changes in global bond demand as explained in the text. It includes our 2026 estimates. Right axis shows the annual change of the yield on the Bloomberg Global Agg index in % (Jan- Oct in dotted lines for 2016 and 2018), and the blue diamond shows the YTD change in 2026.



Source: J.P. Morgan Flows & Liquidity.

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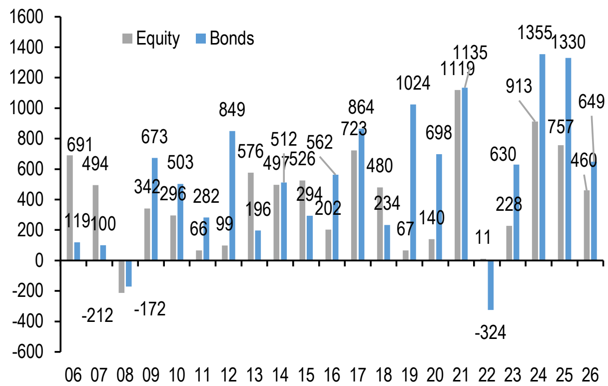
Appendix

Chart A1a: Global equity & bond fund flows

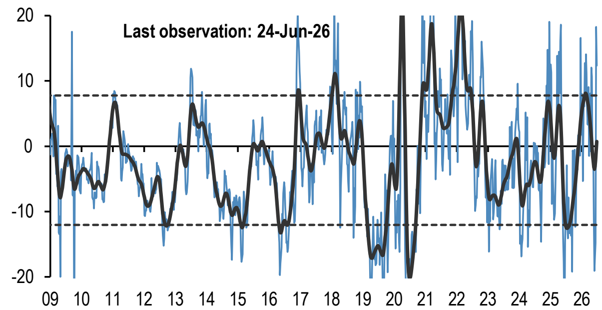
$bn per year of Net Sales, i.e. includes net new sales + reinvested dividends for Mutual Funds and ETFs globally, i.e. for funds domiciled both inside and outside the US. Flows come from ICI (worldwide data up to Q1'26). Data since then are a combination of monthly and weekly data from Lipper and Bloomberg.



Source: ICI, LSEG Lipper, Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

Chart A2: Fund flow indicator

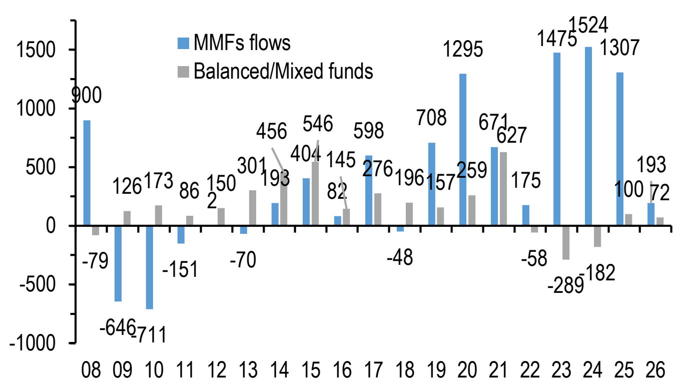
Difference between flows into Equity and Bond funds: $bn per week. Difference between flows into Equity vs. Bond funds in $bn per week. Flows include Mutual Fund and ETF flows globally, i.e. funds domiciled both inside and outside the US (source: Lipper). The thin blue line shows the 4-week average of difference between Equity and Bond fund flows. Dotted lines depict ±1 StDev of the blue line since 2009. The thick black line shows a smoothed version of the same series. The smoothing is done using a Hodrick-Prescott filter with a Lambda parameter of 100. China onshore A-share ETFs have been excluded.



Source: LSEG Lipper, J.P. Morgan Flows & Liquidity.

Chart A1b: Quarterly Balanced/Mixed funds & MMFs flows worldwide

$bn per quarter of Net Sales, i.e. includes net new sales + reinvested dividends for Mutual Funds and ETFs globally, i.e. for funds domiciled both inside and outside the US. Data come from ICI (worldwide data) and are till Q1'26.



Source: ICI, J.P. Morgan Flows & Liquidity.

Table A2: Trading turnover monitor

Volumes are monthly and Turnover ratio is annualised (monthly trading volume annualised divided by the amount outstanding). UST Cash is primary dealer transactions in all US government securities. UST futures are from Bloomberg Finance L.P. JGBs are OTC volumes in all Japanese government securities. Bunds, Gold, Oil and Copper are futures. Gold includes Gold ETFs. Min-Max chart is based on Turnover ratio. For Bunds and Commodities, futures trading volumes are used while the outstanding amount is proxied by open interest. The diamond reflects the latest turnover observation. The thin blue line marks the distance between the min and max for the complete time series since Jan-2005 onwards. Y/Y change is change in YTD notional volumes over the same period last year.

As of Apr-26	MIN	MAX	Turnover ratio	Vol (tr)	y/y chng
Equities					
EM Equity			1.1	$1.7	118%
DM Equity			1.4	$11.4	38%
Govt Bonds					
UST cash			12.9	$20.3	1%
UST futures			0.9	$13.8	3%
JGBs			47.2	¥4,982	8%
Bund futures			1.7	€7.5	12%
Credit					
US HG			1.2	$0.8	11%
US HY			0.6	$0.1	2%
US Convertibles			3.4	$0.04	13%
Commodities					
Gold			65.4	$1.5	67%
Oil			95.4	$3.4	-5%
Copper			2.4	$0.7	80%
Digital Assets					
CME Bitcoin			116.0	$0.075	-34%
CME Ethereum			122.4	$0.036	61%

* Data with one month lag

Source: Bloomberg Finance L.P., Federal Reserve, Trace, Japan Securities Dealer Association, WFE, J.P. Morgan Flows & Liquidity.

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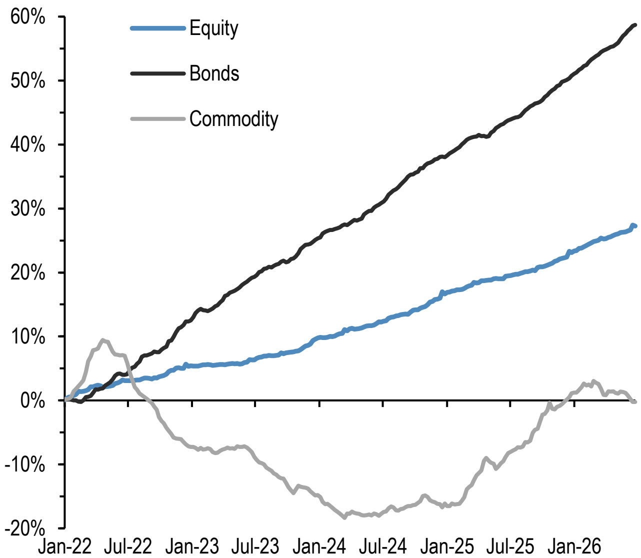
ETF Flow Monitor (as of 24th June)

Chart A3: Global Cross Asset ETF Flows

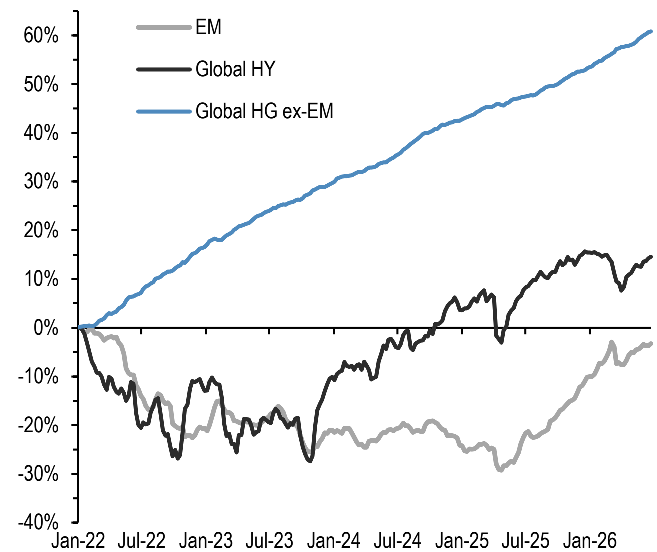
Cumulative flows into ETFs as a % of AUM



Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity. China onshore A-share ETFs have been excluded.

Chart A4: Bond ETF Flows

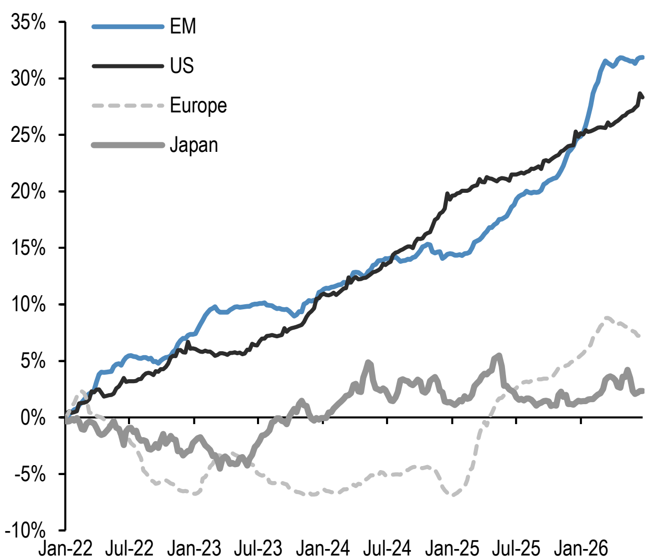
Cumulative flows into bond ETFs as a % of AUM



Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

Chart A5: Global Equity ETF Flows

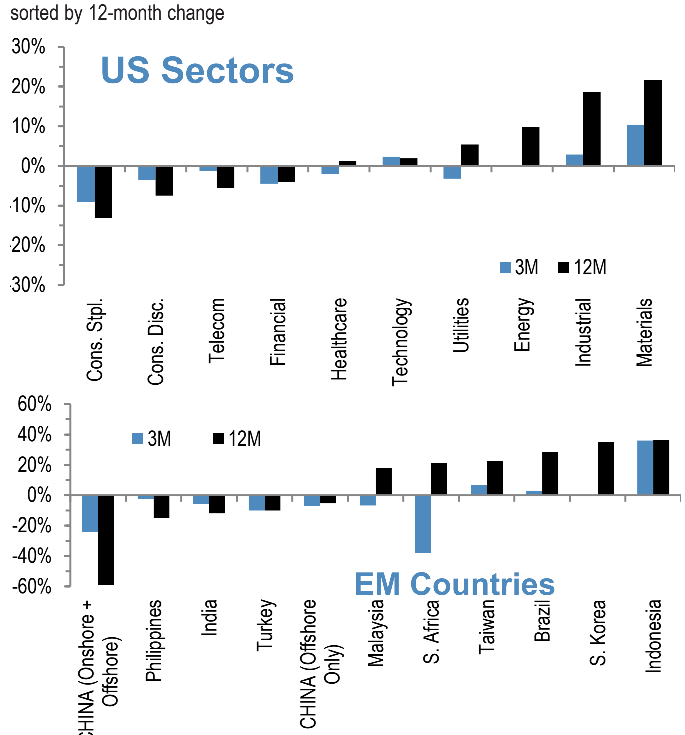
Cumulative flows into global equity ETFs as a % of AUM



Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.
Note: We include ETFs with AUM > $200mn in all the flow monitor charts. In Chart A5 for Japan, we subtract the BoJ buying of ETFs. China onshore A-share ETFs have been excluded.

Chart A6a: Equity Sectoral and Regional ETF Flows

Rolling 3-month and 12-month change in cumulative flows as a % of AUM. Both sorted by 12-month change



Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

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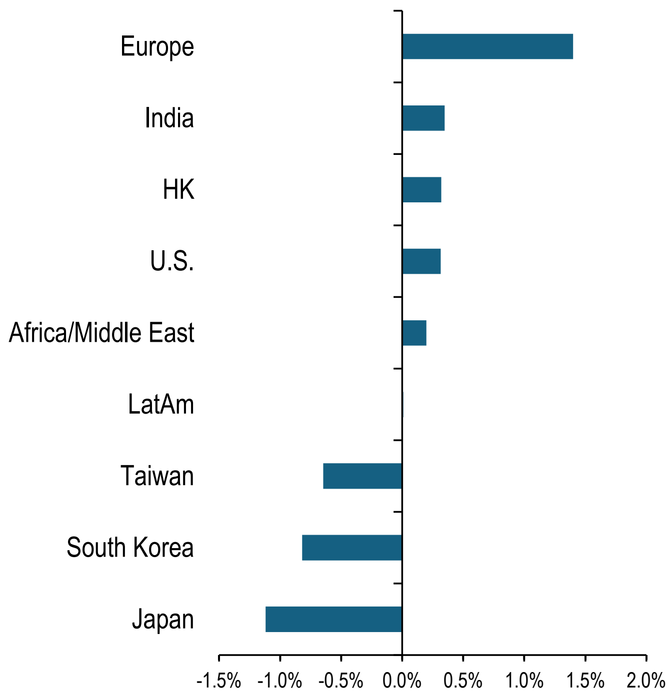
Chart A6b: Regional Equity Allocation Monitor Implied by ETFs

In %.



Source: Bloomberg Finance L.P., MSCI and J.P. Morgan Flows & Liquidity.
Note: Our regional equity allocation monitor implied by ETFs is based on a comparison of the share of each region in the equity ETF space relative to their corresponding shares in global equity indices.

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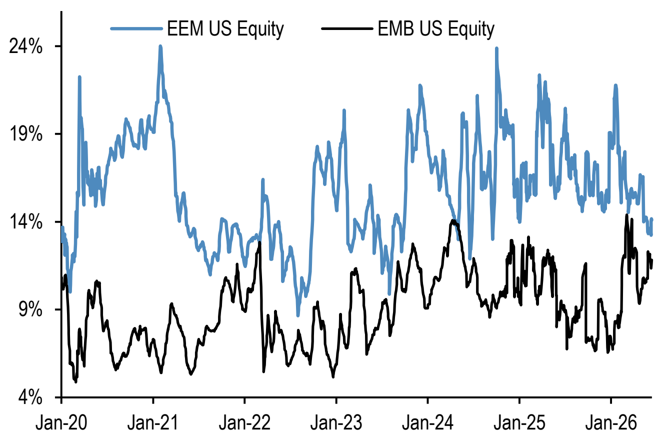
Short Interest Monitor

Chart A7: Short interest on the EEM and EMB US ETF

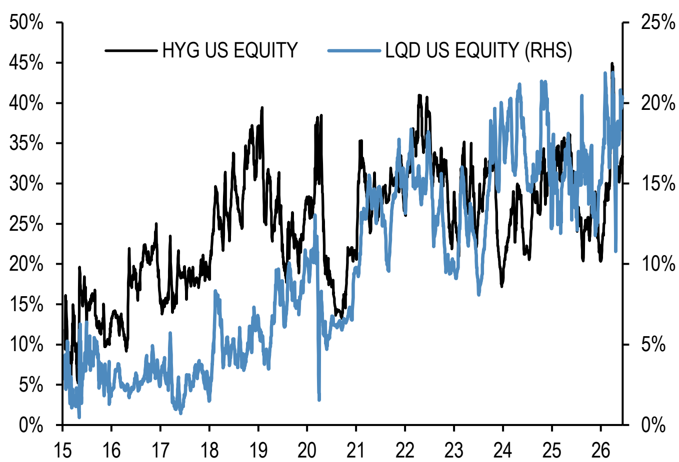
Short Interest as a % share of shares outstanding.



Source: S3, J.P. Morgan Flows & Liquidity.

Chart A8: Short interest on the LQD and HYG US ETF

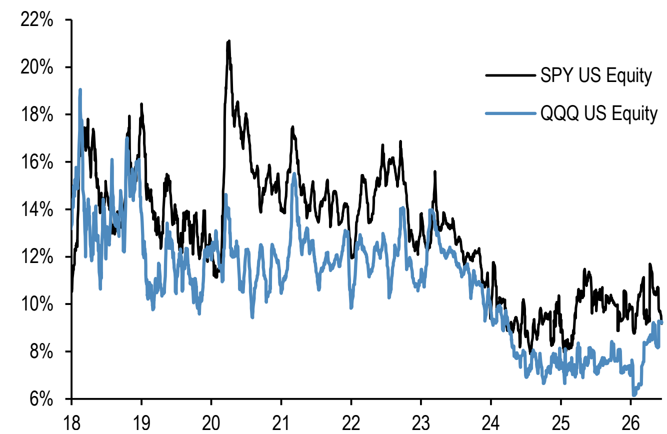
Short Interest as a % share of shares outstanding.



Source: S3, J.P. Morgan Flows & Liquidity.

Chart A9: Short interest on the SPY and QQQ US ETF

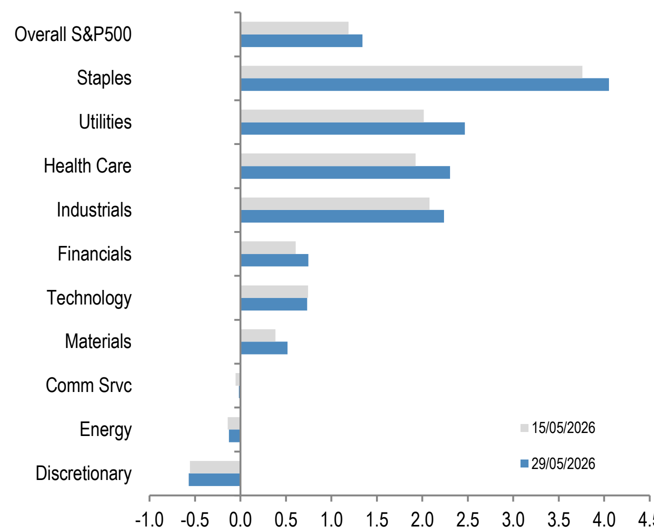
Short Interest as a % share of shares outstanding. Last obs is for 18th June 2026.



Source: S3, J.P. Morgan Flows & Liquidity.

Chart A10: S&P500 sector short interest

Short interest as a % of shares outstanding based on z-scores. A strategy which overweights the S&P500 sectors with the highest short interest z-score (as % of shares o/s) vs. those with the lowest, produced an information ratio of 0.7 with a success rate of 56% (see F&L, Jun 28,2013 for more details).



Source: NYSE, Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

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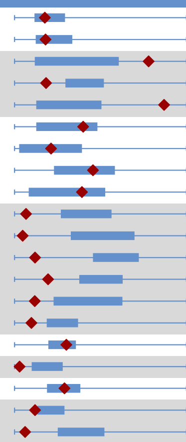
Chart A11a: Cross Asset Volatility Monitor 3m ATM Implied Volatility (1y history), as of 15th June-2026

This table shows the richness/cheapness of current three-month implied volatility levels (red dot) against their one-year historical range (thin blue bar) and the ratio to current realised volatility. Assets with implied volatility outside their 25th/75th percentile range (thick blue bar) are highlighted. The implied-to-realised volatility ratio uses 3-month implied volatilities and 1-month (around 21 trading days) realised volatilities for each asset.

Asset	Current	Low	Low date	High	High date	Upside	Downside	Implied/realized volatility
S&P 500	15%	13%	28-Aug-25	24%	27-Mar-26	9%	2%	0.91x
EuroSTOXX	16%	13%	18-Sep-25	26%	27-Mar-26	11%	2%	0.96x
Nikkei 225	30%	18%	22-Jul-25	33%	31-Mar-26	4%	12%	0.85x
Hang Seng	19%	17%	23-Dec-25	24%	26-Mar-26	6%	1%	0.99x
MSCI EM	34%	12%	15-Aug-25	37%	10-Jun-26	3%	21%	0.87x
Gold	22%	14%	19-Aug-25	33%	26-Mar-26	12%	8%	0.78x
Oil (brent)	40%	27%	19-Sep-25	89%	30-Mar-26	49%	13%	0.78x
Copper	24%	14%	18-Aug-25	37%	03-Feb-26	12%	10%	0.86x
BB commodity index	18%	13%	10-Dec-25	25%	30-Mar-26	7%	5%	1.21x
EUR/USD	9%	9%	29-May-26	13%	27-Mar-26	4%	0%	1.93x
USD/NOK	8%	8%	17-Dec-25	11%	30-Mar-26	3%	0%	1.29x
USD/JPY	7%	7%	29-May-26	11%	01-Jul-25	3%	0%	3.36x
GBP/USD	6%	6%	12-Dec-25	8%	02-Jul-25	2%	0%	1.18x
USD/CHF	7%	6%	29-May-26	9%	01-Jul-25	3%	0%	1.03x
10y US swaps	6%	6%	29-May-26	10%	27-Mar-26	4%	0%	0.98x
10y Eur swaps	59%	43%	26-Feb-26	97%	27-Mar-26	38%	16%	0.87x
CDX IG	39%	38%	29-Jul-25	66%	27-Mar-26	27%	1%	1.03x
CDX HY	39%	31%	09-Jan-26	57%	30-Mar-26	18%	7%	1.08x
iTraxx	41%	37%	09-Jan-26	70%	27-Mar-26	29%	4%	0.89x
iTraxx X/O	20%	19%	23-Dec-25	26%	17-Oct-25	6%	0%	0.44x



Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

Note: Swaps volatility is 3m 10y payer ATMF implied annualized BP vol and credit volatility is 3m 5y on-the-run ATM spread volatility. MSCI EM, Gold, Oil, Copper, BB Commodity Index and Treasury futures are 3m implied vol from Bloomberg.

Definitions:

Current:	Latest available closing level (15-Jun-26)
Low:	Lowest closing level in the last 1y
Low date:	Date the lowest closing level was reached (or the first time it was reached in the case of several identical low closing levels)
High:	Highest closing level in the last 1y
High date:	Date the highest closing level was reached (or the first time it was reached in the case of several identical high closing levels)
Graph:	Shows the current level and the 25th/75th percentile relative to the 1y high/low
Upside:	Implied return/volatility percentage points from current level up to the High (note: return is calculated as simple difference for spread products)
Upside (σ):	Upside in terms of standard deviations (Upside / Current 1y realized volatility)
Downside:	Implied return/volatility percentage points from current level down to the Low (note: return calculated as simple difference for spread products)
Downside (σ):	Downside in terms of standard deviations (Downside / Current 1y realized volatility)
Implied/realized volatility:	Current 3m implied volatility / current realized 3m volatility

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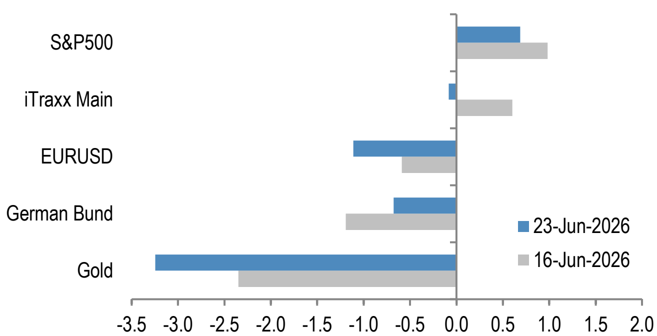
Chart A11b: Option skew monitor

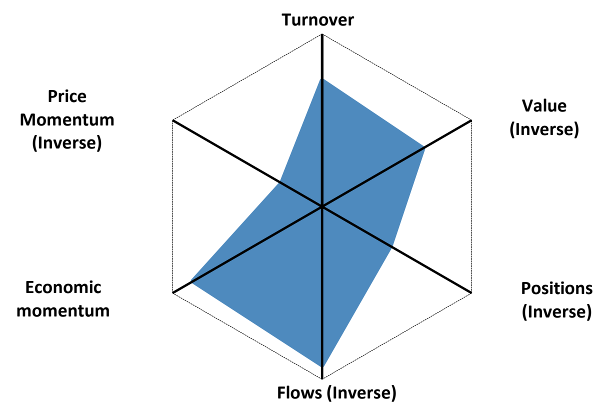
Skew is the difference between the implied volatility of out-of-the-money (OTM) call options and put options. A positive skew implies more demand for calls than puts and a negative skew, higher demand for puts than calls. It can therefore be seen as an indicator of risk perception in that a highly negative skew in equities is indicative of a bearish view. The chart shows z-score of the skew, i.e. the skew minus a rolling 2-year avg skew divided by a rolling two-year standard deviation of the skew. A negative skew on iTraxx Main means investors favour buying protection, i.e. a short risk position. A positive skew for the Bund reflects a long duration view, also a short risk position.



Source: J.P. Morgan Flows & Liquidity.

Chart A11c: Equity market health map

Each of the six axes corresponds to a key indicator for markets. The position of the blue line on each axis shows how far the current observation is from the extremes at either end of the scale. Overall, the larger the blue area within the hexagon, the better the outlook for the S&P500. All variables are expressed as the percentile of the distribution that the observation falls into. I.e. a reading in the middle of the axis means that the observation falls exactly at the median of all historical observations. **Turnover**: Percentile rank of the 21-day rolling S&P500 turnover since 2007. **Value**: Inverse of percentile rank of our quarterly S&P 500 Equity Risk Premium (ERP) estimate since 1957 as described in "A fair value model for US Bonds, Credit and Equities", Panigirtzoglou and Loeys, Jun 2005. **Positions**: Percentile rank of the difference between government bonds and equity positioning from the cross-asset positioning monitor on the front page of our publication since 2016. **Flows**: Inverse of percentile rank of the difference between equity and bond fund flows from the cross-asset fund flow monitor on the front page of our publication since 2007. **Economic momentum**: Percentile rank of the 2-month change in the global manufacturing PMI since 1998, see "REVISITING: Using the Global PMI as trading signal", Nikolaos Panigirtzoglou, Jan 2012). **Price momentum**: Inverse of the percentile rank of the z-score of the difference in the momentum signal of S&P500 vs. 10y USTs since 2016 (based on the average z-score of Short- and Long-term momentum signal in our Trend Following Strategy framework shown in Tables A3 and A4 in the Appendix). China onshore A-share ETFs have been excluded. Last obs. as of 12th June.



Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

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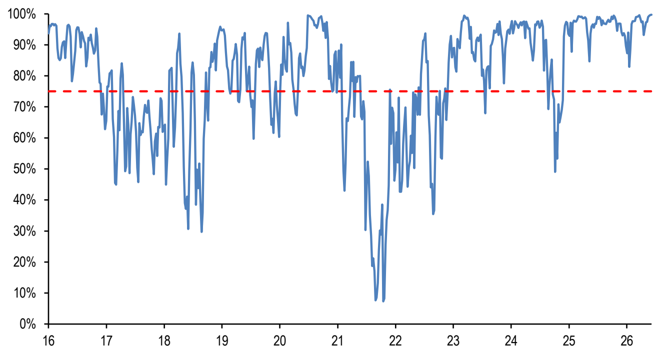
Chart A11d: Implied probability of the S&P500 rising over the next six months with a cutoff set at 75%



Source: J.P. Morgan Flows & Liquidity.

Chart A11e: Logistic regression model for direction of the S&P 500 over a 6 month horizon & Ex-ante probability of an up-move from the logistic regression algorithm vs. the ex-post realised 6-month return in the S&P 500

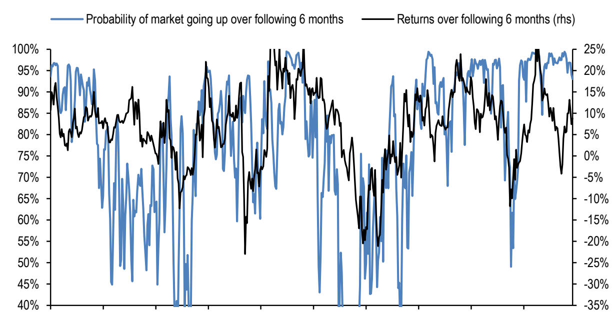
In %.

Dependent Variable: S&500 UP (1) or DOWN (0) over the next six months
Method: ML - Binary Logit (Newton-Raphson / Marquardt steps)

Sample (adjusted): 1/08/2016 12/20/2024
Included observations: 468 after adjustments

Coefficient covariance computed using observed Hessian

Variable	Coefficient	Std. Error	z-Statistic	Prob.
C	3.42	0.84	4.08	0.00
TURNOVER	2.97	0.72	4.15	0.00
VALUE	-3.47	0.90	-3.88	0.00
POSITIONS	-0.77	0.55	-1.39	0.16
FLOWS	-1.79	0.53	-3.38	0.00
ECON_MOMENTUM	3.35	0.62	5.42	0.00
MOMENTUM	-1.36	0.62	-2.19	0.03



Source: J.P. Morgan Flows & Liquidity.

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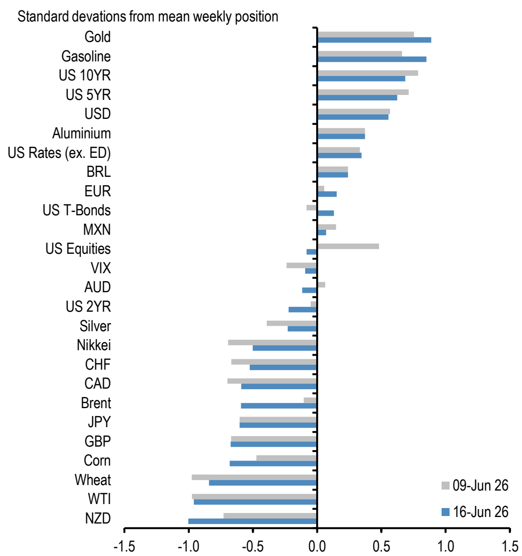
Spec position monitor

Chart A12: Weekly Spec Position Monitor

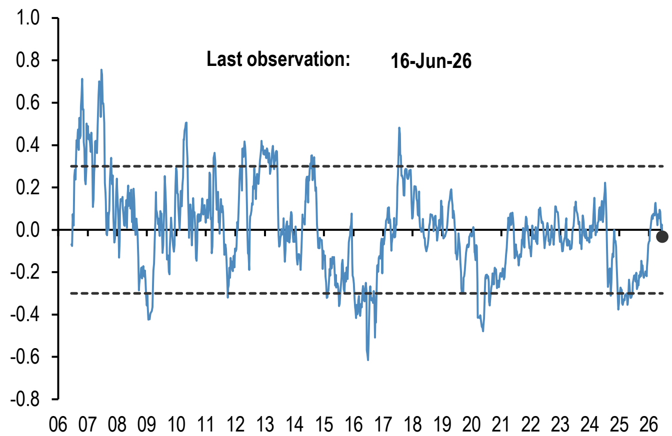
Net spec positions are proxied by the number of long contracts minus the number of short contracts using the speculative category of the Commitments of Traders reports (as reported by CFTC). To proxy for speculative investors for equity and US Treasury bond futures positions we use Asset managers and leveraged funds (see Chart A13), whereas for other assets we use the legacy Non-Commercial category. This net position is then converted to a dollar amount by multiplying by the contract size and then the corresponding futures price. We then scale the net positions by open interest. The chart shows the z-score of these net positions. US rates is a duration-weighted composite of the individual UST futures contracts excluding the Eurodollar contract.



Source: Bloomberg Finance L.P., CFTC, J.P. Morgan Flows & Liquidity.

Chart A14: Spec position indicator on Risky vs. Safe currencies

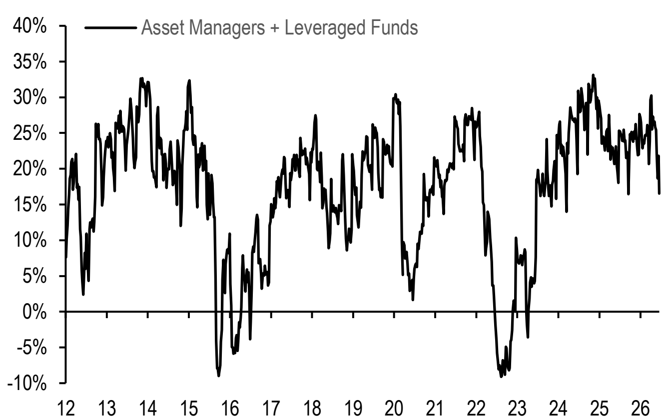
Difference between net spec positions on risky & safe currencies. Net spec position is calculated in USD across 5 'risky' and 3 'safe' currencies (safe currencies also include Gold). These positions are then scaled by open interest and we take an average of 'risky' and 'safe' assets to create two series. The chart is then simply the difference between the"risky" and "safe" series. The final series shown in the chart below is demeaned using data since 2006. The risky currencies are: AUD, NZD,CAD, MXN and BRL. The safe currencies are: JPY, CHF and Gold.



Source: Bloomberg Finance L.P., CFTC, J.P. Morgan Flows & Liquidity.

Chart A13: Positions in US equity futures by Asset managers and Leveraged funds

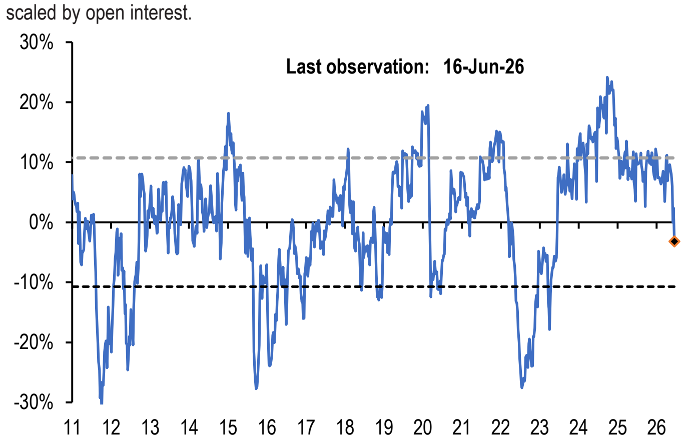
CFTC positions in US equity futures by Leveraged funds and Asset managers (as a % of open interest). It is an aggregate of the S&P500, DowJones, NASDAQ and their Mini futures contracts.



Source: CFTC, Bloomberg Finance L.P. and J.P. Morgan Flows & Liquidity.

Chart A15: Spec position indicator on US equity futures vs. intermediate sector UST futures

Difference between net spec positions on US equity futures vs. intermediate sector UST futures. This indicator is derived by the difference between total CFTC positions in US equity futures by Asset managers + Leveraged Funds scaled by open interest minus the Asset managers + Leveraged Funds spec position on intermediate sector UST futures (i.e. all UST futures duration weighted ex ED and ex 2Y UST futures) also scaled by open interest.



Source: CFTC, Bloomberg Finance L.P. and J.P. Morgan Flows & Liquidity.

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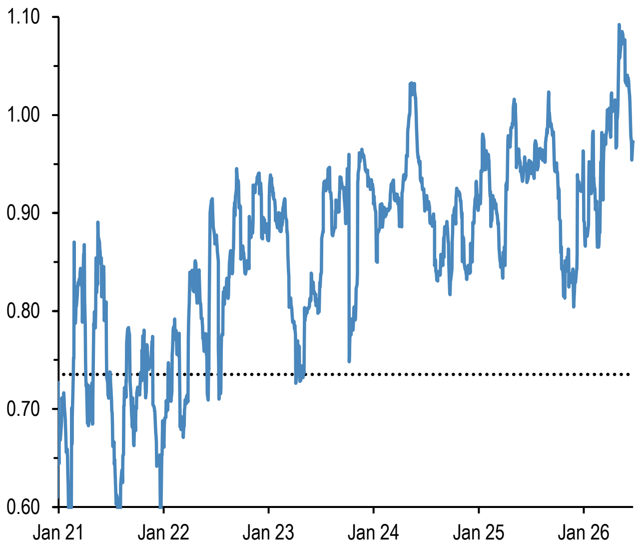
Mutual fund and hedge fund betas

Chart A16: 21-day rolling beta of 20 biggest active US bond mutual fund managers with respect to the US Agg Bond Index

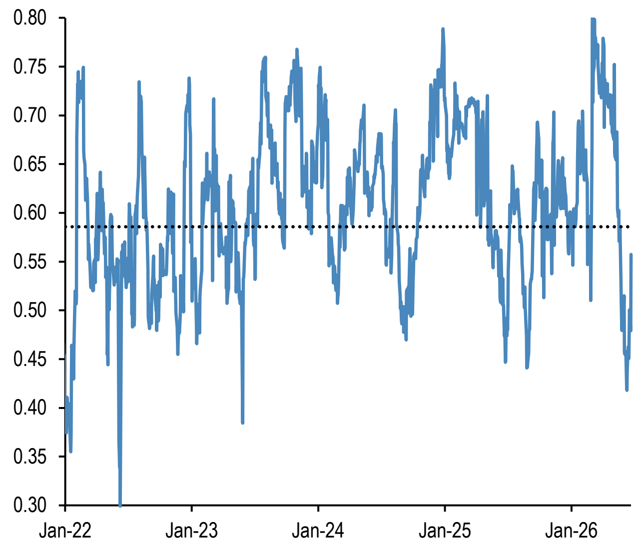
The dotted line shows the average beta since 2013.



Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

Chart A17: 21-day rolling beta of 20 biggest active Euro bond mutual fund managers with respect to the Euro Agg Bond Index

The dotted line shows the average beta since 2013.



Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

Chart A18: Performance of various types of investors

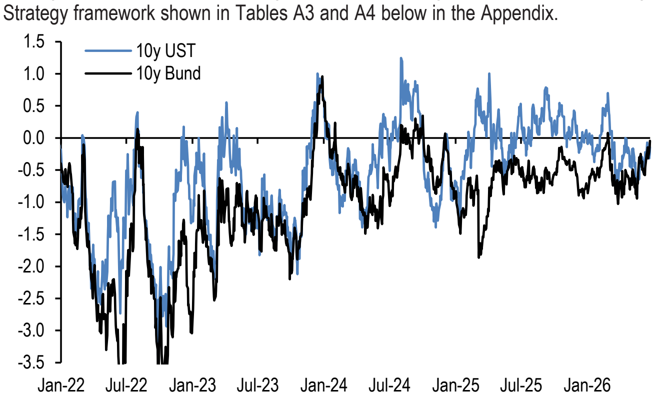
The table depicts the performance of various types of investors in % as of 15th June 2025.

Date	2020	2021	2022	2023	2024	2025	2026
Investors							
Equity L/S	13.4%	9.3%	-7.0%	11.3%	13.0%	15.5%	8.3%
Macro ex-CTAs	4.9%	4.3%	11.5%	1.3%	6.0%	10.2%	4.7%
CTAs	1.2%	8.8%	14.7%	-2.5%	2.9%	-1.1%	8.2%
Risk Parity Funds	3.5%	4.7%	-18.6%	6.0%	5.0%	17.3%	7.2%
US Balanced MFs	13.2%	14.4%	-13.0%	13.8%	11.4%	14.4%	6.9%
Benchmark							
MSCI AC World	16.3%	16.4%	-18.4%	22.2%	17.5%	22.3%	12.3%
Barclays Global Agg	5.6%	-2.5%	-11.2%	7.1%	3.4%	4.9%	0.9%
S&P Riskparity Vol 10	11.5%	12.8%	-16.2%	15.0%	5.4%	15.9%	13.8%
60 US Equity : 40 US Bonds	13.3%	14.8%	-15.4%	18.6%	16.4%	12.7%	7.1%

Source: Bloomberg Finance L.P., HFR, Pivotal Path, J.P. Morgan Flows & Liquidity.

Chart A19: Momentum signals for 10Y UST and 10Y Bunds

Average z-score of Short- and Long-term momentum signal in our Trend Following Strategy framework shown in Tables A3 and A4 below in the Appendix.



Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

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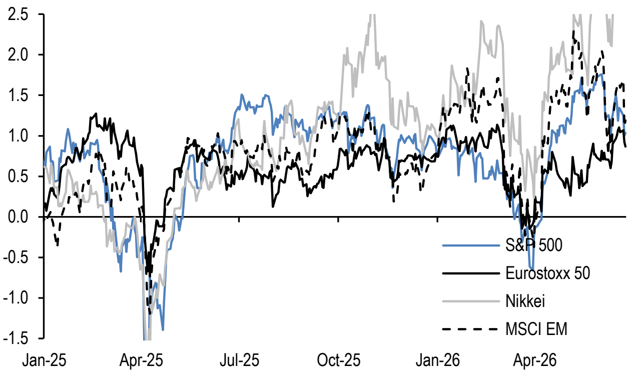
Chart A20: Momentum signals for S&P500

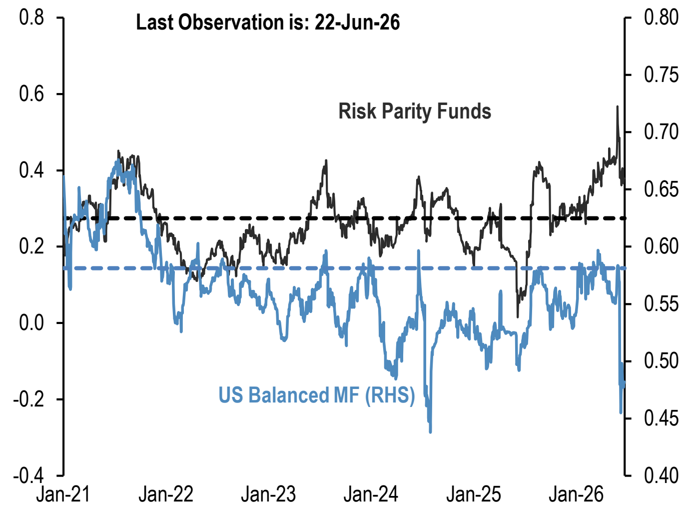
Average z-score of Short- and Long-term momentum signal in our Trend Following Strategy framework shown in Tables A3 and A4 below in the Appendix.



Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

Chart A21: Equity beta of US Balanced Mutual funds and Risk Parity funds

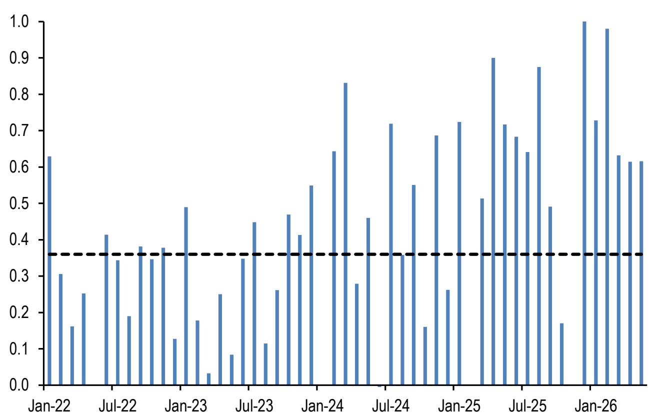
Rolling 42-day equity beta based on a bivariate regression of the daily returns of our Balanced Mutual fund and Risk Parity fund return indices to the daily returns of the S&P 500 and BarCap US Agg indices. Given that these funds invest in both equities and bonds we believe that the bivariate regression will be more suitable for these funds. Our risk parity index consists of 40 daily reporting Risk Parity funds. Our Balanced Mutual fund index includes the top 20 US-based active funds by assets and that have existed since 2006. Our Balanced Mutual fund index has a total AUM of $700bn, which is around half of the total AUM of $1.5tr of US based Balanced funds which we believe to be a good proxy of the overall industry It excludes tracker funds and funds with a low tracking error. Dotted lines are average since 2015.



Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

Chart A22: Equity beta of monthly reporting Equity Long/Short hedge funds

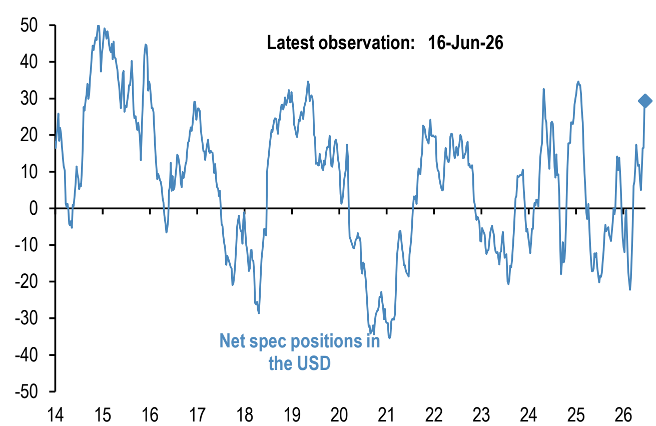
Proxied by the ratio of the monthly performance of Pivotal Path Asset-Weighted Equity Diversified hedge fund index divided by the monthly performance of MSCI AC World Index. Latest obs. is May 26 based on provisional data.



Source: Bloomberg Finance L.P., Pivotal Path, J.P. Morgan Flows & Liquidity.

Chart A23: USD exposure of currency hedge funds

The net spec position in the USD as reported by the CFTC. Spec is the non-commercial category from the CFTC.



Source: CFTC, Barclays, Datastream, Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

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CTAs – Trend following investors' momentum indicators

Table A3: Simple return momentum trading rules across various commodities

Optimal look-back period of each momentum strategy combined with a mean reversion indicator that turns signal neutral when momentum z-score more than 1.5 standard deviations above or below mean, and a filter that turns neutral when the z-score is low (below 0.05 and above -0.05) to avoid excessive trading. Look-backs, current signals and z-scores are shown for shorter-term and longer-term momentum separately, along with performance of a combined signal. Annualised return, volatility and information ratio of the signal; current signal; and z-score of the current return over the relevant look-back period; data from 1999 onward.

		Lookback (moving avg, days)	Annualized return (%)	Vol (%)	IR	Current signal	Time since last change (days)	Z-score	% Change of return index from its moving average
WTI	short	21	5.9	22.7	0.26	0	6	-2.2	-13.6%
	long	462				1	7	0.8	23.9%
Brent	short	84	4.3	21.5	0.20	-1	7	-1.2	-14.3%
	long	504				1	7	0.9	25.8%
Unleaded gas	short	105	2.8	23.2	0.12	1	19	0.3	4.0%
	long	483				0	4	1.5	41.0%
Heat Oil	short	63	4.6	21.9	0.21	-1	12	-1.3	-13.4%
	long	483				0	80	1.8	54.3%
Gasoil	short	63	8.4	21.2	0.40	0	3	-1.6	-17.1%
	long	378				0	80	1.6	46.2%
Nat gas	short	105	13.1	35.9	0.36	-1	12	-0.3	-4.8%
	long	315				-1	101	-0.5	-16.1%
EU emission allowances	short	42	10.1	28.9	0.35	1	9	0.5	4.9%
	long	483				1	9	0.2	6.8%
1m fwd TTF Nat gas	short	21	35.6	31.4	1.14	-1	7	-0.9	-8.9%
	long	294				1	66	0.4	17.9%
Gold	short	21	3.3	10.6	0.31	0	1	-1.8	-5.0%
	long	483				1	12	0.8	10.2%
Silver	short	10	6.6	19.7	0.33	0	0	-2.0	-6.9%
	long	462				1	3	0.9	22.2%
Palladium	short	42	12.3	23.1	0.53	-1	7	-1.4	-11.7%
	long	273				-1	25	-0.7	-16.4%
Platinum	short	105	4.3	18.6	0.23	0	3	-2.0	-18.6%
	long	273				-1	3	-0.5	-8.3%
Aluminium	short	105	5.7	15.2	0.38	-1	1	-0.6	-4.6%
	long	357				1	6	0.9	12.4%
Copper	short	147	7.2	17.3	0.42	1	93	0.3	3.2%
	long	399				1	99	1.0	21.6%
Lead	short	126	0.8	19.2	0.04	-1	11	-0.3	-3.6%
	long	357				-1	112	-0.4	-7.5%
Nickel	short	42	11.9	23.0	0.52	-1	13	-1.0	-8.6%
	long	336				1	125	0.1	2.2%
Zinc	short	126	7.9	19.6	0.40	1	61	0.3	3.8%
	long	399				1	191	0.7	15.7%
Wheat	short	168	4.0	23.2	0.17	0	0	0.0	0.6%
	long	294				0	0	0.0	0.7%
Kansas wheat	short	147	8.0	20.8	0.39	1	24	0.1	1.3%
	long	483				-1	0	-0.1	-1.8%
Corn	short	63	7.9	16.5	0.48	-1	23	-1.3	-9.8%
	long	399				-1	35	-0.8	-14.9%
Soybeans	short	42	4.8	14.5	0.33	-1	17	-0.8	-4.1%
	long	231				-1	3	-0.1	-1.0%
Cotton	short	168	5.5	18.5	0.30	1	8	0.3	4.2%
	long	483				0	4	0.0	-1.0%
Sugar	short	63	6.5	21.6	0.30	-1	20	-0.7	-6.7%
	long	252				-1	28	-0.6	-10.9%
Coffee	short	63	7.8	23.9	0.33	1	0	0.1	0.8%
	long	273				-1	102	-0.5	-10.1%
Cocoa*		10	5.0	31.1	0.16	0	1	3.2	12.0%

* For cocoa, uses only short-term momentum and a z-score threshold of 3 rather than 1.5 as for other contracts.

Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

Table A4: Simple return momentum trading rules across international equity indices, bond futures and FX

Optimal look-back period of each momentum strategy combined with a mean reversion indicator that turns signal neutral when momentum z-score more than 1.5 standard deviations above or below mean, and a filter that turns neutral when the z-score is low (below 0.05 and above -0.05) to avoid excessive trading. Look-backs, current signals and z-scores are shown for shorter-term and longer-term momentum separately, along with performance of a combined signal. Annualised return, volatility and information ratio of the signal; current signal; and z-score of the current return over the relevant look-back period; data from 1999 onward.

		Lookback (moving avg, days)	Annualized return (%)	Vol (%)	IR	Current signal	Time since last change (days)	Z-score	% Change of return index from its moving average
S&P 500	short	84	7.6	12.3	0.62	1	0	0.8	4.1%
	long	315				1	0	1.2	12.2%
Nasdaq 100	short	84	10.2	16.2	0.63	1	0	1.1	8.5%
	long	462				1	0	1.4	26.8%
Nikkei	short	63	4.1	13.3	0.31	0	0	2.3	13.6%
	long	420				0	0	3.5	52.4%
FTSE 100	short	168	5.3	12.1	0.43	1	0	0.6	3.6%
	long	504				0	0	1.6	17.7%
Eurostoxx 50	short	168	4.9	16.6	0.29	1	0	0.9	7.5%
	long	294				1	0	1.0	11.6%
MSCI EM	short	42	12.9	11.5	1.11	1	0	0.4	2.0%
	long	357				0	0	1.8	29.2%
HYG credit	short	168	0.4	3.4	0.13	1	0	0.3	0.8%
	long	273				1	0	0.3	0.9%
LQD credit	short	10	2.6	6.5	0.40	0	0	0.0	0.1%
	long	189				-1	0	-0.1	-0.4%
2Y USTs	short	252	0.9	1.1	0.79	-1	0	-1.0	-1.1%
	long	462				-1	0	-0.6	-1.2%
5Y USTs	short	252	1.5	2.8	0.54	-1	0	-0.8	-1.8%
	long	420				-1	0	-0.4	-1.5%
10Y USTs	short	42	2.0	3.6	0.55	-1	0	-0.2	-0.3%
	long	441				-1	0	-0.3	-1.4%
2Y Schatz	short	189	0.6	0.8	0.80	-1	0	-0.7	-0.5%
	long	504				-1	0	-1.2	-2.1%
5y Bobl	short	63	1.3	1.9	0.71	1	0	0.5	0.5%
	long	483				-1	0	-0.7	-2.3%
10y Bund	short	105	2.1	3.4	0.62	1	0	0.2	0.4%
	long	462				-1	0	-0.4	-2.3%
10Y JGB	short	126	0.9	2.3	0.37	-1	0	-1.2	-1.2%
	long	273				0	0	-2.1	-3.2%
10Y Gilts	short	42	2.2	4.1	0.55	1	0	1.3	2.1%
	long	399				-1	0	-0.4	-2.3%
10y OAT vs Bund	short	189	0.4	1.5	0.27	1	0	0.3	0.3%
	long	294				1	0	0.6	0.6%
10y BTP vs. Bund	short	84	2.3	6.3	0.36	1	0	0.2	0.4%
	long	273				1	0	0.2	0.7%
Euro	short	42	3.4	6.4	0.53	-1	0	-1.2	-2.5%
	long	273				-1	0	-0.9	-5.3%
Yen	short	21	3.6	6.3	0.58	-1	0	-0.7	-1.1%
	long	462				-1	0	-1.5	-12.3%
Sterling	short	168	1.4	7.1	0.20	-1	0	-0.7	-2.8%
	long	294				-1	0	-0.7	-3.9%
AUD	short	42	3.8	7.8	0.49	-1	0	-1.2	-3.3%
	long	420				1	0	0.1	1.2%
CAD	short	168	1.5	5.9	0.25	-1	0	-1.2	-4.6%
	long	504				-1	0	-1.2	-7.8%
G10 FX	short	42	2.2	5.2	0.42	1	0	1.4	2.3%
	long	189				1	0	0.8	2.8%
EMCI	short	42	2.6	5.0	0.52	-1	0	-0.4	-0.7%
	long	189				1	0	0.5	2.2%

Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

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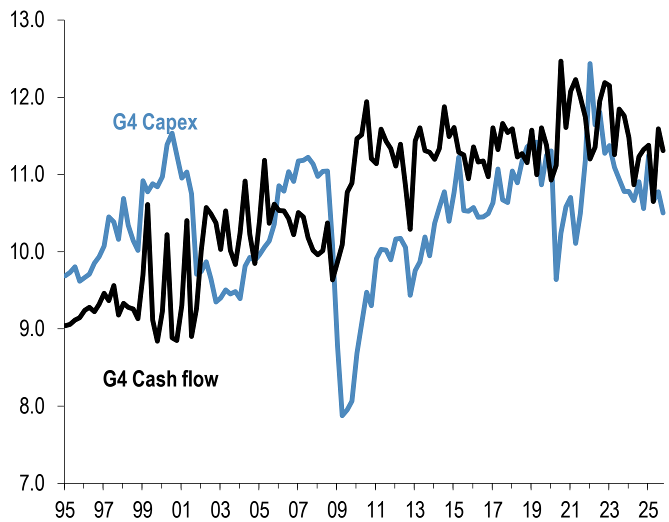
Corporate Activity

Chart A24: G4 non-financial corporate capex and cash flow as % of GDP

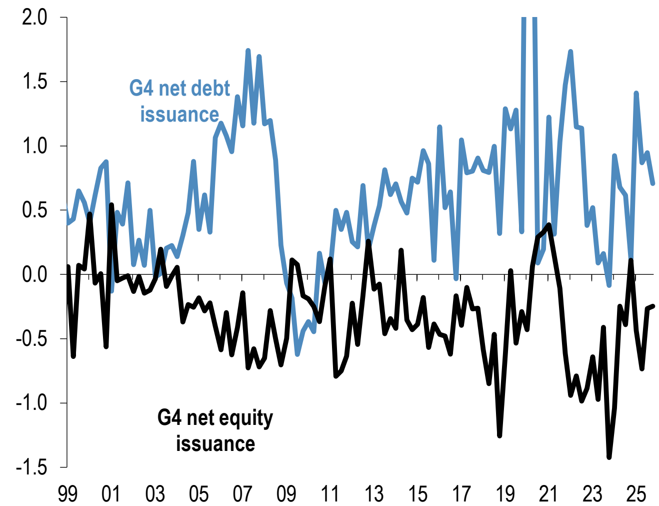
% of GDP, G4 includes the US, the UK, the Euro area and Japan. Last observation as of Q4 2025.



Source: ECB, BOJ, BOE, Federal Reserve flow of funds, J.P. Morgan Flows & Liquidity.

Chart A25: G4 non-financial corporate sector net debt and equity issuance

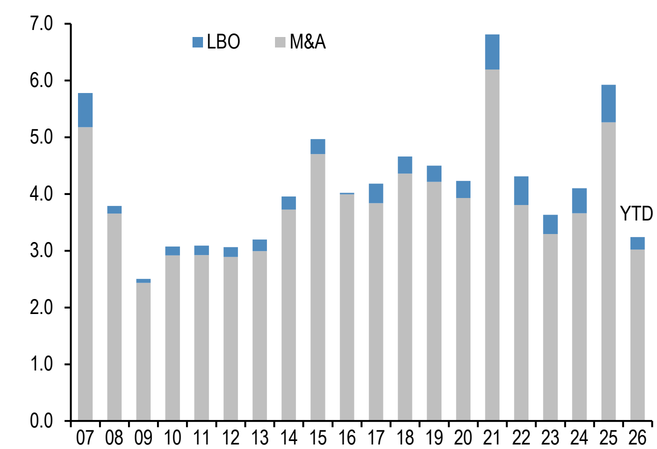
$tr per quarter, G4 includes the US, the UK, the Euro area and Japan. Last observation as of Q4 2025.



Source: ECB, BOJ, BOE, Federal Reserve flow of funds, J.P. Morgan Flows & Liquidity.

Chart A26: Global M&A and LBO

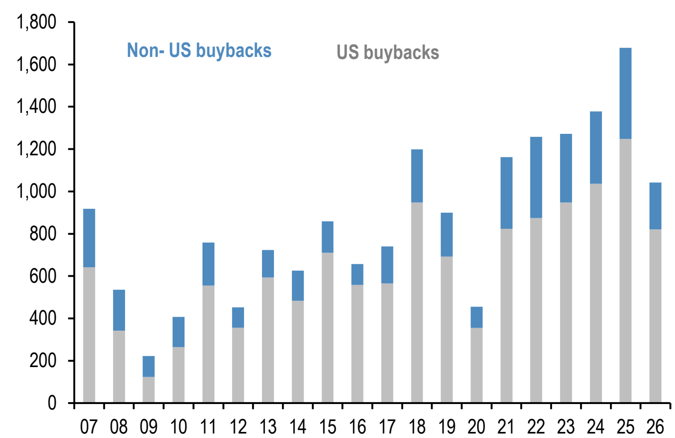
$tr. M&A and LBOs are announced.



Source: Dealogic, J.P. Morgan Flows & Liquidity.

Chart A27: US and non-US share buyback

$bn, are as of May 26. Buybacks are as announced.



Source: Bloomberg Finance L.P., LSEG, J.P. Morgan Flows & Liquidity.

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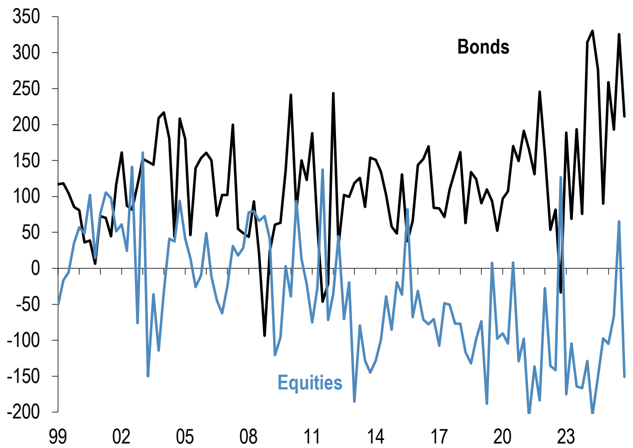
Pension fund and insurance company flows

Chart A28: G4 pension funds and insurance companies equity and bond flows

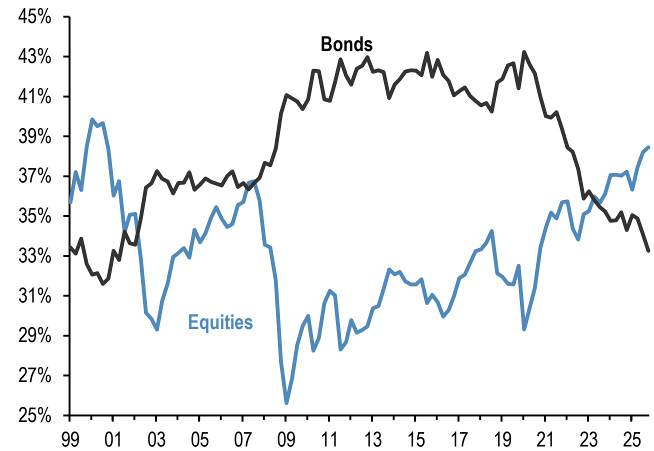
Equity and bond buying in $bn per quarter. G4 includes the US, the UK, Euro area and Japan. Last observation is Q4 2025.



Source: ECB, BOJ, BOE, Federal Reserve flow of funds, J.P. Morgan Flows & Liquidity.

Chart A29: G4 pension funds and insurance companies equity and bond levels

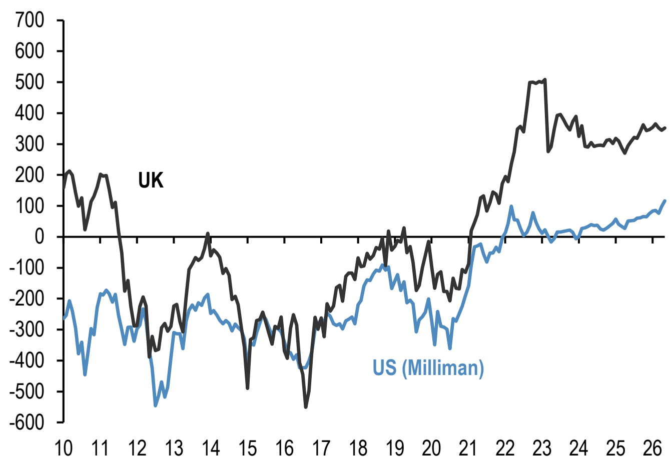
Equity and bond as % of total assets per quarter. G4 includes the US, the UK, Euro area and Japan. Last observation is Q4 2025.



Source: ECB, BOJ, BOE, Federal Reserve flow of funds, J.P. Morgan Flows & Liquidity.

Chart A30: Pension fund deficits

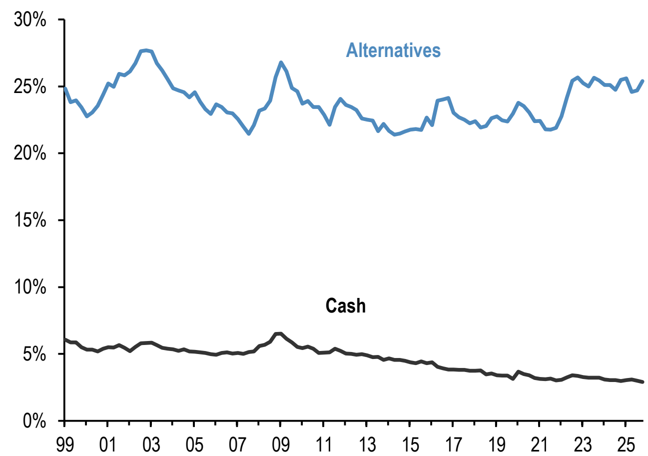
US$bn. For US, funded status of the 100 largest corporate defined benefit pension plans, from Milliman. For UK, funded status of the defined benefit schemes eligible for entry to the Pension Protection Fund, converted to US$ at today's exchange rates. Last obs. is May'26 for US & UK.



Source: Milliman, UK Pension Protection Fund, J.P. Morgan Flows & Liquidity.

Chart A31: G4 pension funds and insurance companies cash and alternatives levels

Cash and alternative investments as % of total assets per quarter. G4 includes the US, the UK, Euro area and Japan. Last observation is Q4 2025.



Source: ECB, BOJ, BOE, Federal Reserve flow of funds, J.P. Morgan Flows & Liquidity.

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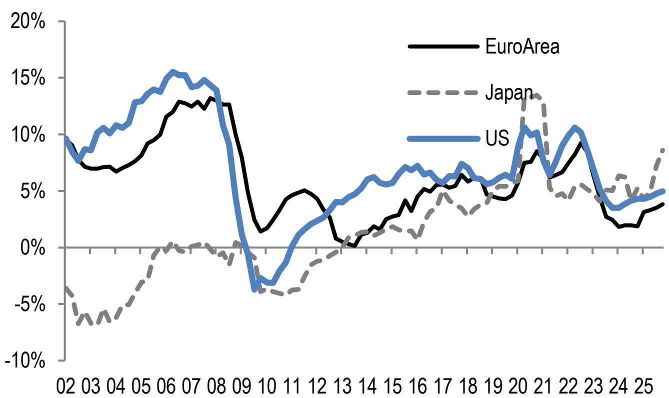
Credit Creation

Chart A32: Credit creation in the US, Japan and Euro area

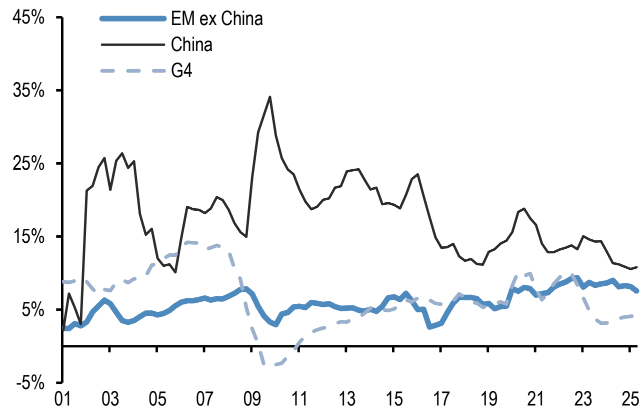
Rolling sum of 4-quarter credit creation as % of GDP. Credit creation includes both bank loans as well as net debt issuance by non-financial corporations and households. Last obs. is Q4'25 for Japan, Euro Area and US.



Source: Fed, ECB, BoJ, Bloomberg Finance L.P., and J.P. Morgan Flows & Liquidity.

Chart A33: Credit creation in EM

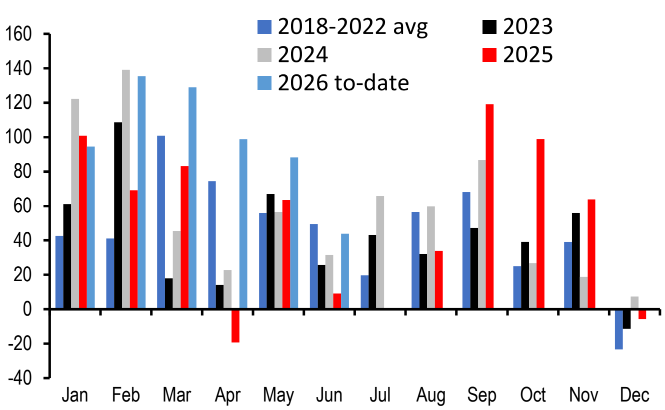
Rolling sum of 4-quarter credit creation as % of GDP. Credit creation includes both bank loans as well as net debt issuance by non-financial corporations and households. The chart uses the FX rates as of the latest observation to calculate the credit creation at constant exchange rates. Last obs. is for Q2'25.



Source: G4 Central banks FoF, BIS, ICI, Barcap, Bloomberg Finance L.P., IMF, and J.P. Morgan Flows & Liquidity.

Chart A34: Monthly net issuance of US HG bonds

$bn. Up to June 11th.



Source: Dealogic, J.P. Morgan Flows & Liquidity.

Table A5: Equity and Bond issuance

$bn, Equity supply and corporate announcements are based on announced deals, not completed. M&A is announced deal value and buybacks are announced transactions. Y/Y change is change in YTD announcements over the same period last year.

Equity Supply	12-Jun	4 wk avg	13 wk avg	y/y chng
Global IPOs	4.0	3.9	4.7	86%
Secondary Offerings	3.3	16.6	11.0	28%
Corporate announcements				
M&A - Global	79.8	133.3	124.1	36%
- US Target	27.9	92.3	60.3	60%
- Non-US Target	51.9	41.0	63.8	19%
Net bond issuance	**Feb-26**	**3 mth avg**	**YTD avg**	**y/y chng**
USD	8	144	179	-50%
Non-USD	162	129	277	-4%

Source: Bloomberg Finance L.P., Dealogic, Thomson Reuters, J.P. Morgan Flows & Liquidity.

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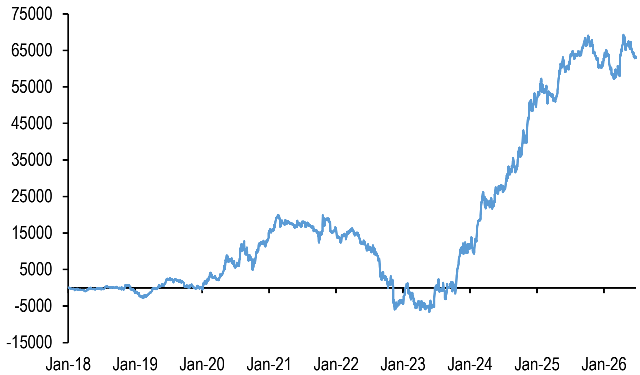
Bitcoin monitor

Chart A35: Our Bitcoin position proxy, based on open interest in CME Bitcoin futures contracts

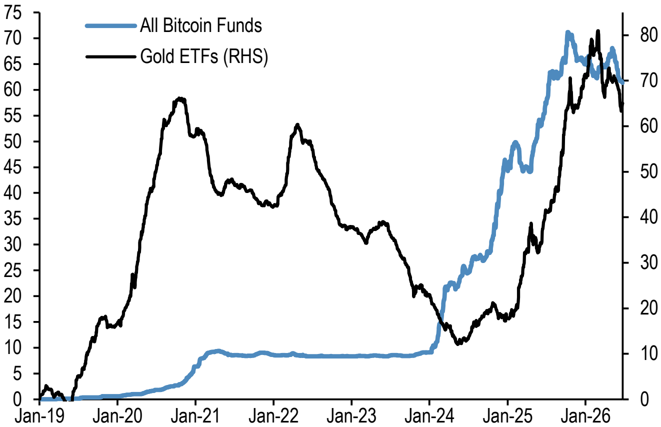
In number of contracts. Last obs. for 23rd June 2026.



Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

Chart A36: Cumulative Flows in all Bitcoin funds and Gold ETF holdings

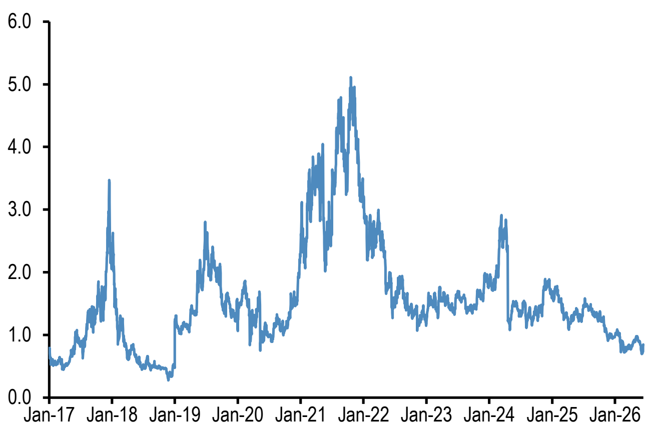
Both the y-axis in $bn.



Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

Chart A37: Ratio of bitcoin market price to production cost

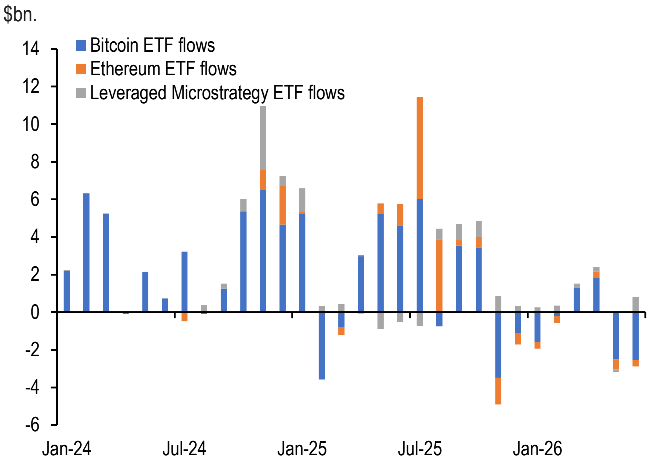
Based on the cost of production approach following Hayes (2018).



Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

Chart A38: Monthly flows into spot bitcoin ETFs, spot ethereum ETFs and leveraged MicroStrategy ETF

$bn.



Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

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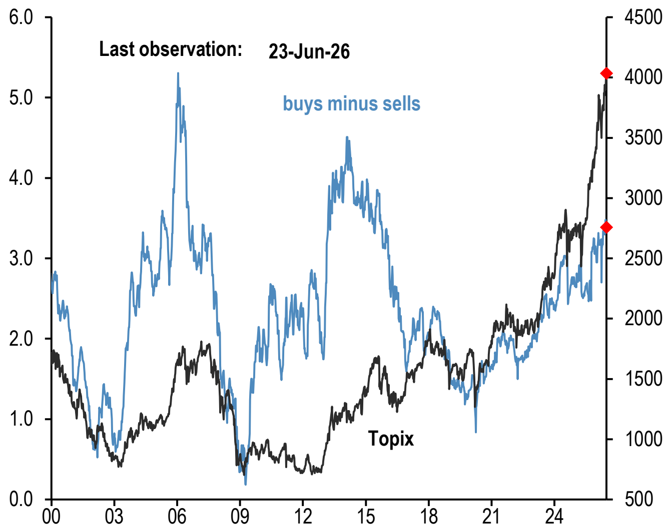
Japanese flows and positions

Chart A39: Tokyo Stock Exchange margin trading: total buys minus total sells

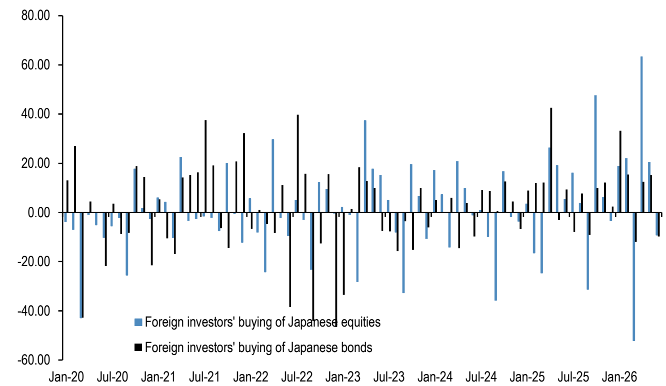
In bn of shares. Topix on right axis.



Source: Tokyo Stock Exchange, Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

Chart A40: Monthly net purchases of Japanese bonds and Japanese equities by foreign residents

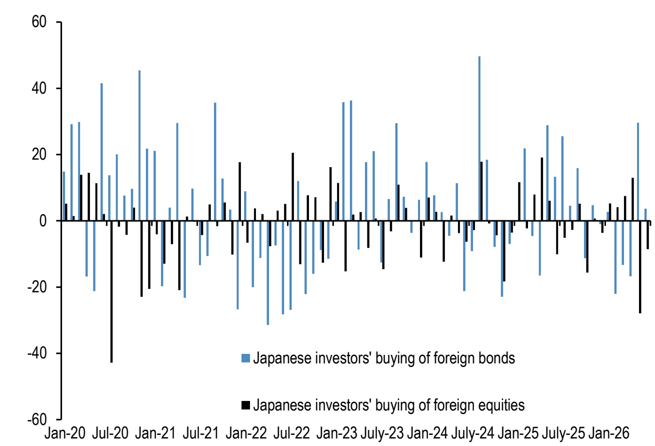
$bn, Last weekly obs. is for 12th June' 26.



Source: Japan MoF, Bloomberg Finance L.P., and J.P. Morgan Flows & Liquidity.

Chart A41: Monthly net purchases of foreign bonds and foreign equities by Japanese residents

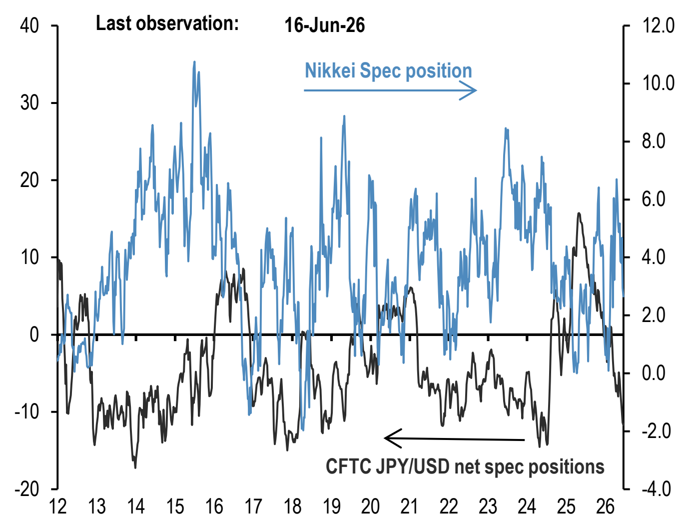
$bn, Last weekly obs. is for 12thJune' 26.



Source: Japan MoF, Bloomberg Finance L.P., and J.P. Morgan Flows & Liquidity.

Chart A42: Overseas CFTC spec positions

CFTC spec positions are in $bn. For Nikkei we use CFTC positions in Nikkei futures (USD & JPY) by Leveraged funds and Asset managers.



Source: Bloomberg Finance L.P., CFTC, J.P. Morgan Flows & Liquidity.

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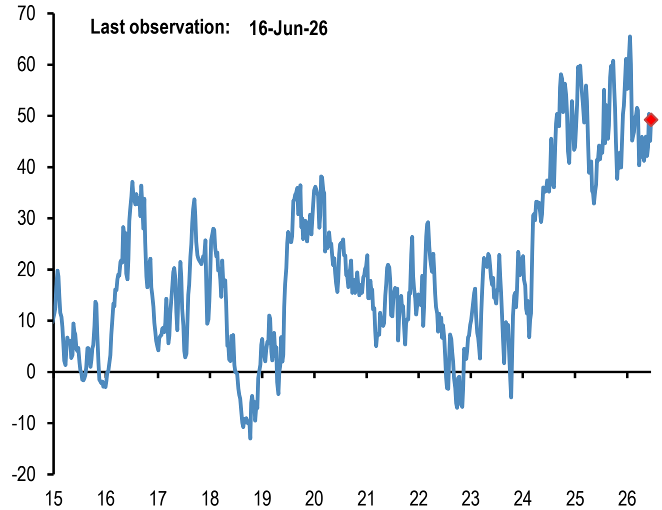
Commodity flows and positions

Chart A43: Gold spec positions

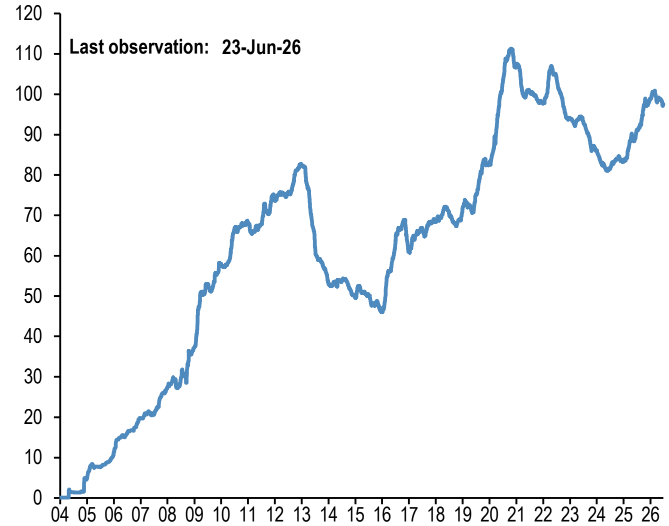
$bn. CFTC net long minus short position in futures for the Managed Money category.



Source: CFTC, Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

Chart A44: Gold ETFs

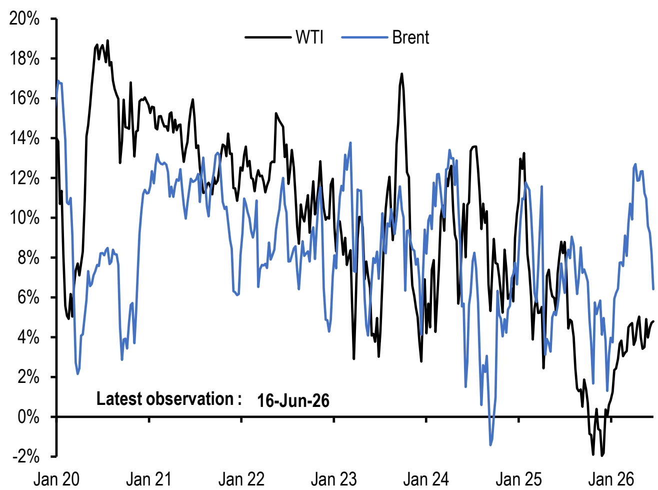
Mn troy oz. Physical gold held by all gold ETFs globally.



Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

Chart A45: Oil spec positions

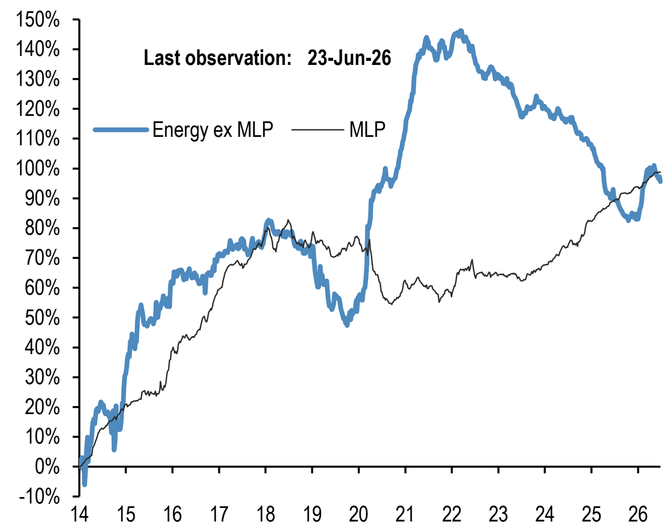
Net spec positions divided by open interest. CFTC futures positions for WTI and Brent are net long minus short for the Managed Money category.



Source: CFTC, Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

Chart A46: Energy ETF flows

Cumulative energy ETFs flows as a % of AUM. MLP refers to the Alerian MLP ETF.



Source: CFTC, Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

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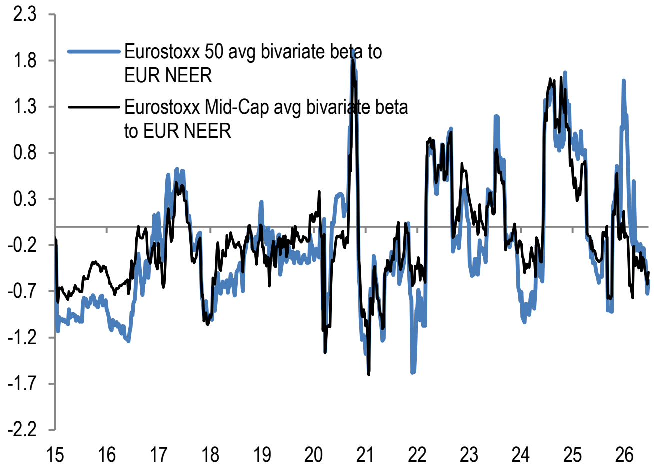
Corporate FX hedging proxies

Chart A47: Average beta of Eurostoxx 50 companies and Eurostoxx Mid-Cap to trade-weighted EUR

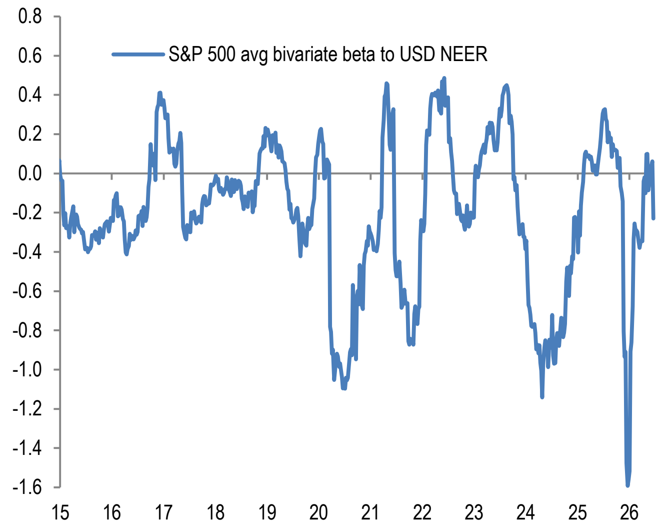
Rolling 26 weeks average betas based on a bivariate regression of the weekly returns of individual stocks in the Eurostoxx 50 index to the weekly returns of the MSCI AC World and JPM EUR Nominal broad effective exchange rate (NEER).



Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

Chart A48: Average beta of S&P500 companies to trade-weighted US dollar

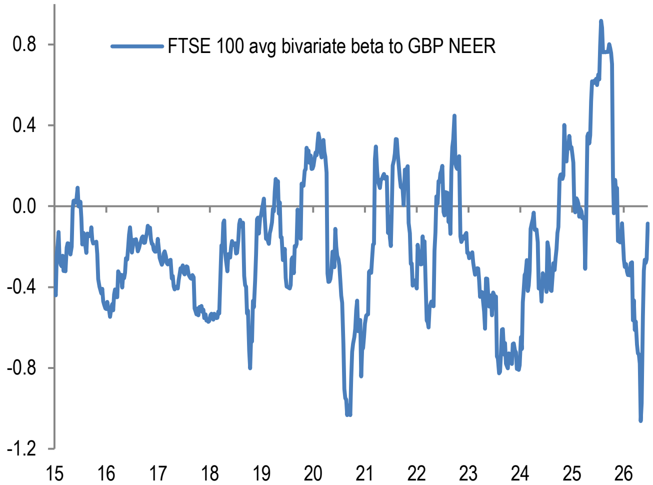
Rolling 26 weeks average betas based on a bivariate regression of the weekly returns of stocks in the S&P500 index to the weekly returns of the MSCI AC World and JPM USD Nominal broad effective exchange rate(NEER).



Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

Chart A49: Average beta of FTSE 100 companies to trade-weighted GBP

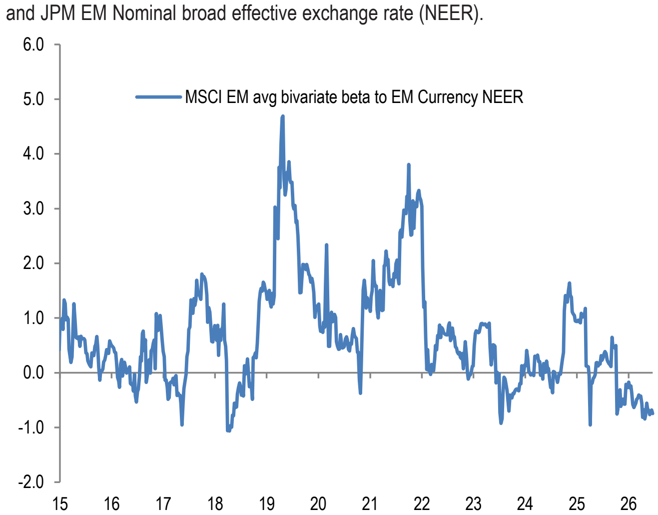
Rolling 26 weeks average betas based on a bivariate regression of the weekly returns of individual stocks in the FTSE 100 index to the weekly returns of the MSCI AC World and JPM GBP Nominal broad effective exchange rate (NEER).



Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

Chart A50: Average beta of MSCI EM companies to trade-weighted EM Currency Index

Rolling 26 weeks average betas based on a bivariate regression of the weekly returns of individual stocks in the MSCI EM index to the weekly returns of the MSCI AC World and JPM EM Nominal broad effective exchange rate (NEER).



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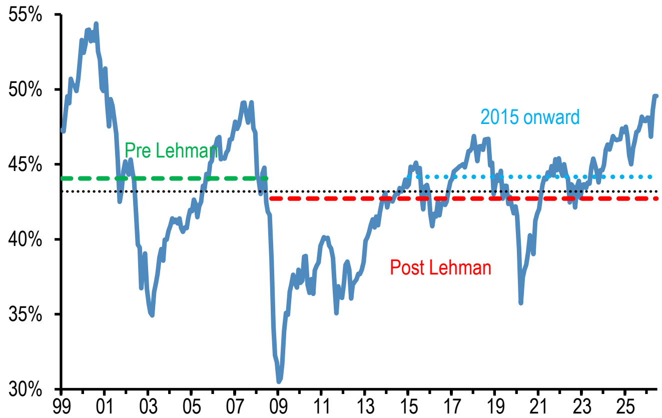
Non-Bank investors' implied allocations

Chart A51: Implied equity allocation by non-bank investors globally

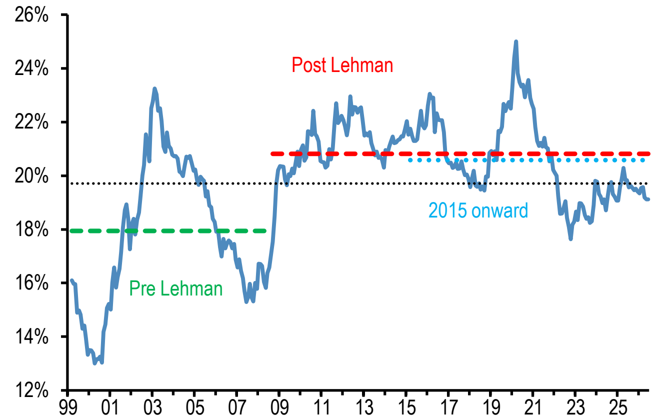
Global equities as % total holdings of equities/bonds/M2 by non-bank investors/ Commodities including gold held by private investors. Dotted lines are averages.



Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

Chart A52: Implied bond allocation by non-bank investors globally

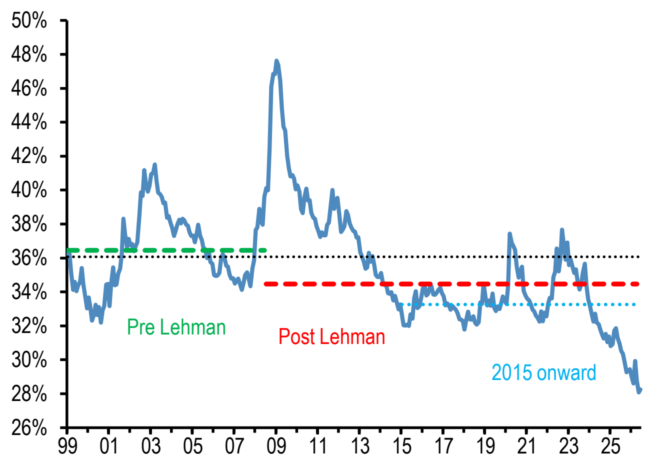
Global bonds as % total holdings of equities/bonds/M2 by non-bank investors/ Commodities including gold held by private investors. Dotted lines are averages.



Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

Chart A53: Implied cash allocation by non-bank investors globally

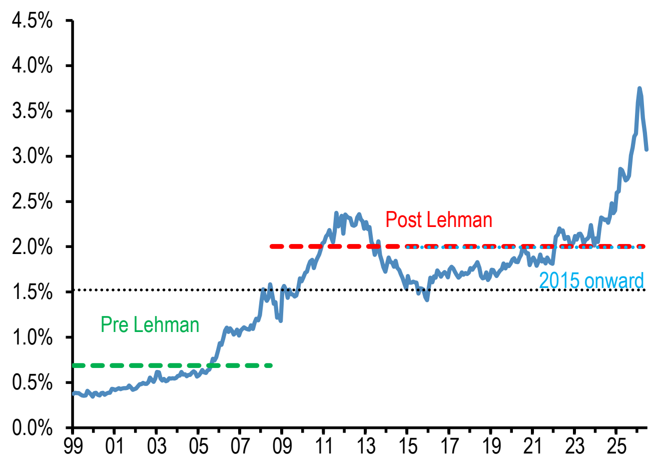
Global cash held by non-bank investors as % total holdings of equities/bonds/M2 by non-bank investors/Commodities including gold held by private investors. Dotted lines are averages.



Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

Chart A54: Implied commodity allocation (including privately held gold) by non-bank investors globally

Proxied by the open interest of commodity futures ex gold and privately held gold as % of the stock of equities, bonds, cash held by non-bank investors globally and commodities including gold held by private investors.



Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

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